Closing the Security Activation Gap: A Security-by-Design Methodology for IoT Ecosystems and Its Empirical Validation in a DC Smart Microgrid

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Abstract

A persistent security activation gap constrains IoT adoption: security mechanisms are widely available in commercial platforms yet remain inactive after deployment. We present a security-by-design methodology for centralized IoT ecosystems that treats identity as an intrinsic entity property, spans seven stages across four phases, and integrates cybersecurity throughout the development lifecycle. A dynamic Root of Trust process establishes verifiable cryptographic identity for each entity, the foundation for authentication, access control, and secure communication.

We validate the methodology through a confirmatory case study applying all seven stages to a DC Smart Microgrid: two ESP32 nodes and a Raspberry Pi 4B coordination agent. Four case study questions assess guidance suffi-

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ciency, identity model unambiguity, Root of Trust verifiability, and threat coverage. All seven stages produced documented, traceable artifacts with no external guidance required; a 14-parameter identity model across four entity types was produced without ambiguity; the methodology achieves unweighted OWASP IoT Top 10 coverage of 6/10 items fully and 4/10 partially addressed. The Root of Trust reconstructs each node's cryptographic identity from a Physically Unclonable Function response rather than a stored key, deriving its session key from an ephemeral ECDH exchange for forward secrecy; it was validated end-to-end on real hardware for both nodes, including a detailed $N = 30$ hard-reset trial (30/30 success, median handshake latency 1300 ms, mean 1313 ± 48 ms) that an independent trial on the second node confirmed generalizes across physical chips. The methodology's parameterized, technology-agnostic structure is designed to generalize across diverse IoT domains; empirical validation in additional domains remains future work.

Keywords: Internet of Things, Security-by-Design, IoT Methodology, Root of Trust, Identity Management, Smart Microgrid, Case Study Research

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1. Introduction

The Internet of Things (IoT) has evolved from a conceptual vision into a technological paradigm underpinning critical infrastructure, industrial automation, healthcare systems, and smart cities (Al-Fuqaha et al., 2015; Zhou et al., 2021). Despite this expansion, large-scale IoT adoption remains constrained by fragmented architectures, heterogeneous technologies, and—critically—the failure to integrate cybersecurity from the earliest design stages (Shahinzadeh et al., 2024; Makhdoom et al., 2018).

The consequences of this gap are well documented. The Mirai botnet, which compromised over 600,000 IoT devices to launch unprecedented distributed denial-of-service attacks, originated from misconfigured or unprotected deployments relying on default credentials (Antonakakis et al., 2017). Subsequent analyses confirm that cyberattacks on IoT infrastructure continue to escalate in frequency and sophistication, targeting energy systems, healthcare networks, and public utilities (Abraham et al., 2025; Okika et al., 2025).

A central observation motivating this work is that the principal barrier to secure IoT deployment is not the *absence* of security features but the *security activation gap*: the disconnect between security capabilities *available*

in a platform and those *operational* after a standard deployment (Figure 1). Widely adopted open-source IoT platforms provide security mechanisms that require manual post-installation configuration, effectively restricting their use to organizations with dedicated security expertise (Fagan et al., 2021).

A second, more fundamental barrier operates at the methodological level: most IoT development projects lack a structured process that integrates cybersecurity from organizational decision-making through operational maintenance. Security is typically addressed *after* architectural and technology decisions are made, producing mechanisms that are bolt-on rather than foundational, with identity management that is platform-specific rather than entity-centric.

To address both barriers, we present two complementary contributions:

1. **A security-by-design methodology** for developing centralized IoT ecosystems. The methodology treats identity as an intrinsic property of every entity, spans seven stages across four phases (organizational strategy, architecture, implementation, and operation), and embeds cybersecurity—from threat policy to Root of Trust establishment—as a foundational concern from problem definition through ongoing operation. A dynamic Root of Trust (RoT) process establishes verifiable cryptographic identity for each entity before any operational communication begins.

2. **An empirical confirmatory case study** applying the methodology to the design and implementation of a DC Smart Microgrid (SMG) IoT ecosystem, following Yin's (2014) Case Study Research design. The SMG comprises two ESP32 secondary control nodes and a Raspberry Pi 4B primary coordination agent, reaching Technology Readiness Level 3. Four case study questions evaluate guidance sufficiency, identity model unambiguity, Root of Trust verifiability, and OWASP IoT threat coverage.

Conceptual foundation: The identity-centric IoT paradigm, the IoT/ecosystem distinction, the formal entity identity model, and the reconceptualization of the Root of Trust as a multi-phase process are established at the conceptual and definitional level in a companion manuscript (Campos-Sanchez et al., 2026a). The present work builds directly on that foundation to specify, validate, and empirically evaluate an operative

seven-stage development methodology; it does not re-derive these definitions independently.

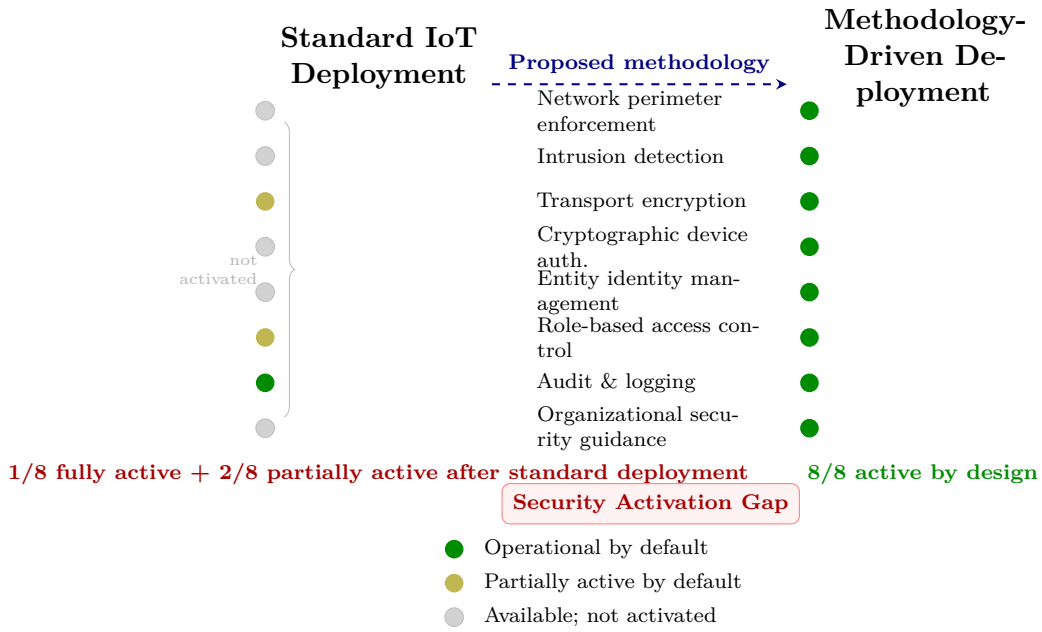


Figure 1: The security activation gap: standard IoT deployments activate only a subset of platform security controls by default; the remaining controls require manual post-installation configuration by practitioners with dedicated security expertise. The proposed methodology specifies activation of all controls by design; the TRL 3 case study reported here realizes this for most but not all (flash encryption, secure boot, and JTAG disable remain deferred to ITER 2/TRL 5+ — Limitations), consistent with an early-maturity proof of concept.

Scope note: The ESP32/C (ESP-IDF) firmware suite and the primary agent software implementing the SMG are presented as part of this work's own case study rather than as a companion contribution under separate review. The present work evaluates the *methodology* through the case study; implementation details are provided only to the extent required for methodological evidence.

We evaluate the methodology through: (a) stage-by-stage traceability, demonstrating that all seven stages produce documented, actionable artifacts; (b) quantitative measurements of identity provisioning and handshake latency; and (c) a systematic OWASP (Open Web Application Security

Project) IoT Top 10 (OWASP Foundation, 2018) coverage analysis.

The remainder of this article is organized as follows. Section 2 reviews related work. Section 3 presents the conceptual framework. Section 4 details the methodology and its STRIDE threat coverage model. Section 5 presents the SMG case study. Section 6 reports empirical results. Section 7 discusses findings, and Section 8 concludes.

2. Related Work

We adopt the definition of IoT methodology from Fortino et al. (2020): a structured set of guidelines, methods, practices, and documents organized within a logical framework for solving the specific problem of IoT ecosystem development. We evaluate existing methodologies against nine criteria: the eight classical dimensions drawn from the literature, supplemented by a ninth, *security validation*, which captures whether the methodology's security provisions are supported by empirical evidence from a formal case study or experiment.

Table 1 compares representative methodologies. A 2023 comparative study (Moreta et al., 2023) confirms cybersecurity integration remains the most underdeveloped dimension across IoT development methodologies: most proposals emphasize functional aspects—interoperability, scalability, technological integration—while treating cybersecurity as an afterthought or omitting it entirely (Fortino et al., 2020; Al-Fuqaha et al., 2015; Merzouk et al., 2020). Agent-based approaches (Manate et al., 2014; Ayala et al., 2015; Spanoudakis and Moraitis, 2015) provide structured development processes but lack explicit security mechanisms and identity assignment; model-driven methodologies (Morin et al., 2017; Costa et al., 2016; Cicirelli et al., 2017) address code heterogeneity and portability but not entity identity or security. A systematic review of agile IoT methodologies (Guerrero-Ulloa et al., 2023) confirms even recent structured approaches defer authentication, identity management, and threat modeling until after functional requirements are met—if addressed at all.

Security-focused proposals remain limited in scope and depth. Josyula and Gupta (2017) presents a security methodology for IoT systems engineering without specific guidelines for identity assignment or cryptographic mechanism selection. Savaglio et al. (2018) includes security aspects (authorization, key management, authentication) but not selection criteria or their

timing relative to development stages. Muñoz et al. (2025) proposes a five-layer service-oriented architecture with security as a cross-cutting concern, without explaining how mechanisms are integrated or selected; its PKI-based scheme neglects resource-constrained devices. A systematic review of IoT authentication schemes in this journal (Saqib and Moon, 2023) (over 120 proposals, 2010–2022) confirms core fragmentation: authentication mechanisms are proposed in isolation, without connection to a broader development process, formal identity model, or measurable threat coverage metric. Blanco et al. (2023) propose an ontology-driven framework for cyber-physical security requirements, providing structured elicitation but not a full lifecycle spanning organizational policy through operational deployment.

More recently, several efforts target specific IoT security domains without holistic development guidance: Zero Trust architectures for industrial IoT (Zanasi et al., 2024; Fernandez and Brazhuk, 2024), blockchain-based supply chain security (Ray et al., 2024; Fernando et al., 2023), and SDN-based IoT fog security (Javanmardi et al., 2023; Turner et al., 2023) address individual threat surfaces but not a development methodology. Visintin (2024) examines identity of things device-centrically but does not extend to ecosystem-wide identity management across heterogeneous entity types or connect identity to access control policy. PUF-based Root of Trust implementations (Sassalou et al., 2025) and hardware security modules (Bakiris and Oikonomakis, 2025) strengthen device-level identity but remain single-layer solutions, leaving the broader development process unaddressed. This work adopts an SRAM-based PUF for the ESP32, `esp32_puflib` (Stanicek, 2022), as the concrete Stage 4 technology for identity-parameter generation without persistent on-device key storage (Section 4), integrated into the full seven-stage process rather than treated as an isolated device-level mechanism.

Standards from government agencies and consortia—Secure-by-Design (UK Government Security, 2024), NIST IoT Cybersecurity Guidance (Fagan et al., 2021), ETSI EN 303 645 (European Telecommunications Standards Institute, 2020), the CSA IoT Security Controls Framework (Cloud Security Alliance, 2023), and the 2025 NIST cybersecurity supply chain risk guidance (Licata et al., 2025)—provide valuable security baselines. However, these standards remain predominantly prescriptive at the policy level, offering no technology selection criteria, no process for deriving implementation choices from organizational constraints, and no deployment automation.

Taken together, these approaches share a common limitation: the absence of a methodology that simultaneously addresses (a) organizational adoption

strategy, (b) technology-agnostic architectural design with a formal entity-centric identity model, (c) a dynamic Root of Trust process establishing verifiable cryptographic identity, (d) iterative development guided by maturity levels, and (e) empirical validation through a formal case study with measurable artifacts at every stage. The methodology presented in this work addresses all five dimensions. The validation follows Yin's (2014) confirmatory case study design with four research questions and documented stage-by-stage artifact traceability—we are not aware of a comparable empirical validation of a complete IoT security development methodology using this research design.

Table 1: Comparison of IoT ecosystem development methodologies (9 criteria). Tech.Dev. = technological development guidance; CyberSec. = explicit cybersecurity integration; Agnostic = technology-agnostic design; Org. = organizational aspects; Scalability; Maturity = TRL or equivalent; Interop. = interoperability; Concept. = formal conceptual framework; Sec.Val. = security provisions validated through formal case study or experiment. ✓ full, ~ partial, ✗ absent. *This work* is rated ~ on Tech.Dev. because the methodology is technology-agnostic by design: it specifies mechanism categories and stage requirements, not specific technology selection; ~ on Maturity and Scalability because the case study reaches only TRL 3 (laboratory proof of concept) on 2 nodes, not a production-scale or externally-validated deployment. Ratings for "This work" are self-assessed by the authors, not externally validated.

Methodology	Tech. Dev.	Cyber Sec.	Agnostic	Org.	Scal-ability	Maturity	Inter-op.	Concept. Frame.	Sec. Val.
Manate et al. (2014) (Manate et al., 2014)	~	~	✗	✗	✓	✗	✓	✗	✗
Ayala et al. (2015) (Ayala et al., 2015)	✓	✗	✓	✗	~	✗	✓	~	✗
Spanoudakis et al. (2015) (Spanoudakis and Moraitis, 2015)	✓	✗	✗	~	✓	✗	✓	✗	✗
Patel et al. (2015) (Patel and Cassou, 2015)	✓	✗	✗	✗	✓	✗	✓	✓	✗
Costa et al. (2016) (Costa et al., 2016)	✓	✗	✗	✗	✓	✗	✓	✗	✗
Fortino et al. (2017) (Fortino et al., 2017)	✓	✗	✗	✗	✓	✓	✓	✗	✗
Josyula et al. (2017) (Josyula and Gupta, 2017)	✗	~	✓	✗	✓	✗	✓	~	✗
Savaglio et al. (2018) (Savaglio et al., 2018)	~	~	✓	✗	✓	✗	✓	~	✗
Casadei et al. (2019) (Casadei et al., 2019)	✓	✗	~	✗	✓	✗	✓	~	✗
Guerrero-Ulloa et al. (2023) (Guerrero-Ulloa et al., 2023)	✓	✗	✓	~	~	✗	~	✗	✗
Muñoz et al. (2025) (Muñoz et al., 2025)	✓	~	✗	✗	~	✗	~	~	✗
This work (2026)	~	✓	✓	✓	~	~	✓	✓	✓

3. Conceptual Framework

Ambiguity in IoT definitions hinders coherent methodology design (Sorri et al., 2022). Rather than re-deriving reference definitions, this methodology builds directly on the identity-centric conceptual framework established in a companion manuscript (Campos-Sanchez et al., 2026a), which reframes IoT as a technological paradigm in which cyber-physical systems are constitutively endowed with identity, and distinguishes this paradigm from an *IoT*

ecosystem—a digital environment in which entities possessing unique identities interact under a defined structure to provide services that solve identified problems. We adopt both definitions unchanged and use them as the entry point for the present methodology.

The ecosystem comprises seven element types: IoT devices (embedded systems with ecosystem-relative identity), IoT servers (network-connected infrastructure enabling entity communication), IoT services (processes enabling ecosystem operation and security), storage systems (information persistence infrastructure), users (business-logic-level entities with defined roles), administrators (operational management entities), and communication infrastructure (data exchange enablers).

Definition 1 (**Cybersecurity in an IoT Ecosystem**). *Intrinsic properties of the design, development, structure, integration, operation, and physical conditions that determine the vulnerability, robustness, resilience, and response to disturbances or malicious interventions in IoT ecosystems.*

We likewise adopt the reconceptualization of the Root of Trust (RoT) as a multi-phase *process*, rather than a static hardware component, established in Campos-Sanchez et al. (2026a): entities establish, exchange, and securely store identity parameters within a trusted environment before any operational communication begins. This process accommodates heterogeneous device capabilities—from constrained microcontrollers to cloud servers—supports both symmetric and asymmetric cryptographic schemes, and establishes the trusted context required for provisioning, authentication, and secure session establishment across all entity types. When Phase 1 (parameter generation) is instantiated through a Physically Unclonable Function (PUF) rather than random generation followed by persistent storage, Phase 2 (secure storage) is correspondingly trivialized: the entity's private identity parameter is never stored at rest—only non-secret helper data required to reconstruct it deterministically—removing the secure-storage attack surface rather than merely hardening it. Stage 6.1.3 (Section 4) instantiates this process concretely for the case study's entity types.

4. Proposed Methodology

The methodology is organized into four phases and seven stages (Figure 2). Each stage produces documented outputs that serve as inputs to the subsequent stage, and is itself decomposed into numbered sub-processes (e.g.,

2.2.4, 6.1.3) that the case-study evidence cites throughout this article; the Supplementary Material gives both the generic, technology-agnostic specification of each process (Tables S1–S4) and its instantiation for the SMG case study (Tables S5–S8).

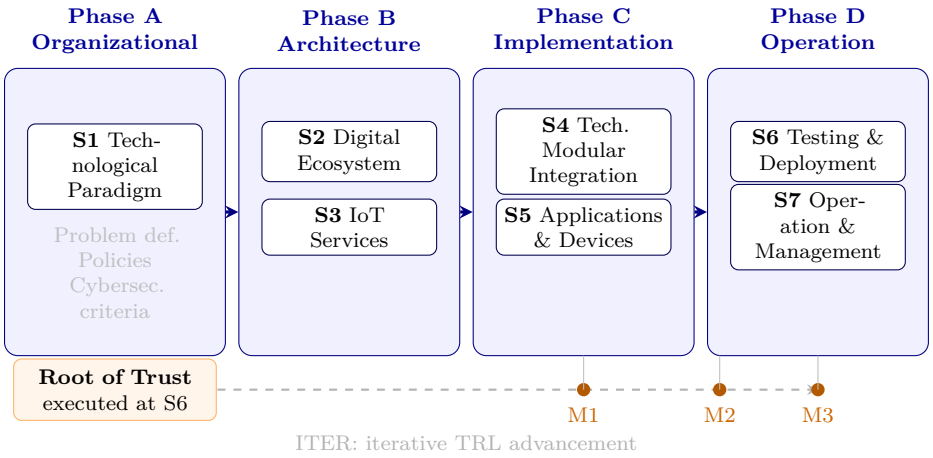


Figure 2: Organization of the proposed methodology: four phases (Organizational, Architecture, Implementation, Operation) comprising seven numbered stages (S1–S7). Iterative development cycles (ITER) advance Technology Readiness Level. Maintenance points M1 (implementation), M2 (deployment), and M3 (operation) trigger targeted stage re-entry. The Root of Trust is executed at Stage 6.

4.1. Phase A: Organizational (Stage 1)

This phase addresses organizational considerations for IoT adoption. Stage 1 establishes the problem to be solved, operating policies, development policies, and cybersecurity criteria. The key processes are: problem definition with cost-benefit analysis (1.1); operating policies covering functional objectives, financial constraints, and regulatory compliance (1.2); development policies defining an iterative strategy using the TRL framework, timelines, and resource allocation (1.3); and cybersecurity policies covering criticality assessment, technological sovereignty, data sovereignty, supply chain security, and applicable government guidelines including Secure-by-Design (UK Government Security, 2024) and Privacy-by-Design (European Union Agency for Cybersecurity, 2017) principles (1.4). Processes 1.4.2–1.4.3 operationalize the organizational-strategy layer of the sovereignty framework established in Campos-Sanchez et al. (2026a), under which technological

and data sovereignty are treated as emergent properties of who controls the entities' Root of Trust process rather than as external policy overlays.

4.2. Phase B: Architecture (Stages 2–3)

Stage 2 defines the IoT ecosystem architecture at the entity level: integration diagram, communication topology, entity hierarchy, functional processes, data flow, and scalability strategy. Critically, Stage 2 instantiates the formal identity model established in Campos-Sanchez et al. (2026a), under which each entity X in ecosystem E possesses an identity:

$$I_X^E = P_{X\,pub}^E \cup P_{X\,priv}^E \tag{1}$$

where public parameters $P_{X\,pub}^E$ serve as identifiers and private parameters $P_{X\,priv}^E$ enable cryptographic operations, and authentication parameters $P_{X\,auth}^E \subset I_X^E$ and shared secrets $P_{X\,SS}^E \subset P_{X\,auth}^E$ are derived from this identity foundation (Figure 3). This methodology extends that model with an explicit cardinality count for each entity type X: the public-parameter and private-parameter counts are:

$$n_X^E = |P_{X\,pub}^E| \tag{2}$$

$$k_X^E = |P_{X\,priv}^E| \tag{3}$$

so that the total identity parameter count is $n_X^E + k_X^E$ and the table columns report the two component counts and their sum.

Stage 3 specifies the services required for entity operation: secure storage subsystems, access control mechanisms, identity parameter storage strategies, and regulatory compliance integration. For constrained devices, Stage 3 requires an explicit cost/risk evaluation of persistent secure-storage options (hardware security modules, encrypted flash partitions) against physically-derived alternatives that avoid persisting a secret altogether; a Physically Unclonable Function (PUF) that regenerates the private identity parameter on demand from non-secret helper data is one such alternative, and its selection at Stage 4 is a direct consequence of this Stage 3 evaluation rather than an independent design choice. Servers and users retain conventional secrets management and MFA as generic Stage-3 options; the SMG case study itself implements only a single-factor password hash for users/admins, not MFA (Table 12, Limitations).

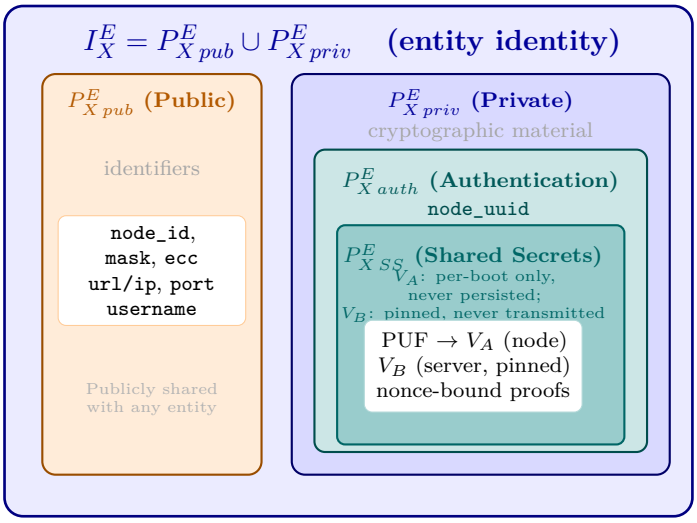


Figure 3: Identity parameter set hierarchy for entity X in ecosystem E. Shared secrets P_{SS} enable cryptographic operations; authentication parameters P_{auth} prove identity; private parameters P_{priv} are never disclosed; public parameters P_{pub} serve as identifiers. In the SMG, the node's private identity parameter (its PUF response) lives in P_{priv} but is never persisted to storage—only the non-secret mask/ECC helper data used to reconstruct it lives in ESP32 NVS, as P_{pub}.

4.3. Phase C: Implementation (Stages 4–5)

Stage 4 selects enabling technologies aligned with policies from Stage 1 and architecture from Stages 2–3. Technology choices are evaluated against maturity level (TRL), vulnerability landscape, and supply chain risk. For the SMG, Stage 4 selects `esp32_puflib` (Stanicek, 2022), an SRAM-based PUF implementation for the ESP32 family, as the identity-generation mechanism motivated by the Stage 3 secure-storage evaluation above. Because `esp32_puflib` is implemented as a native C/ESP-IDF component, Stage 4 also selects a C/ESP-IDF firmware architecture for the node, so that the PUF library is linked directly without a language-runtime wrapper; the same decision selects ESP-IDF's native `mbedtls/esp_aes` for AES-256-GCM and ECDSA support. This platform choice is itself a direct, traceable consequence of the Stage 3 evaluation, not an independent implementation preference. Stage 5 implements all ecosystem elements: servers, devices, databases, communication infrastructure, IoT services, and applications. Security is enforced through secure coding practices, unit testing of security mechanisms, and validation against applicable security guidelines (Secure-by-Design principles (UK Government Security, 2024), NIST IoT Cybersecurity Guidance (Fagan et al., 2021), and OWASP IoT Top 10).

4.4. Phase D: Operation (Stages 6–7)

Stage 6 covers deployment, configuration, and testing: individual component validation, infrastructure deployment, *Root of Trust execution* for all entities (Process 6.1.3), general operation verification, and security testing (penetration testing at higher TRLs; tamper verification at TRL 3). Stage 7 addresses ongoing operation: monitoring, element maintenance, information lifecycle management, threat intelligence, element updates, incident response, and network infrastructure oversight.

4.5. Iterative Development and Maintenance

The methodology supports iterative advancement through TRL-guided cycles (ITER) and three structured maintenance points: M1 (during implementation, triggers technology re-evaluation), M2 (during deployment, triggers code revision), and M3 (during operation, triggers architectural redesign). M1 fires when an implementation-phase review reveals that the chosen technology cannot satisfy a stage requirement; M2 fires when deployment-phase testing reveals that the implementation requires code-level revision to meet a security or functional requirement; M3 fires when operation-phase

monitoring reveals that the deployed architecture cannot accommodate an observed operational pattern. This structure supports progressive maturation from proof-of-concept to production deployment.

4.6. Root of Trust Establishment

Building on the four-phase Root of Trust process of Campos-Sanchez et al. (2026a), we instantiate its parameter generation, secure storage, exchange, and peer-management phases as concrete protocol message sequences for both cryptographic schemes.

PUF-seeded device identity (resource-constrained devices): Rather than generating a random device key pair and persisting the private key in flash, the device's identity key pair is derived on demand from a Physically Unclonable Function (PUF), instantiated via `esp32_puflib` (Stanicek, 2022). *Enrollment* (performed once, Table 2) reads the ESP32's RTC FAST SRAM power-up state across repeated measurements and derives two artifacts—a stable-bit mask and error-correction (helper) data—that are stored on the device and allow the same physical response to be reconstructed, with correction, on every subsequent boot. Reconstructing the response from these artifacts alone, without the chip, is intended to be infeasible, consistent with the design goal of SRAM-based PUFs (Guajardo et al., 2007); however, repetition-code helper data is known to leak some response information (Dodis et al., 2004; Maes, 2013; Delvaux et al., 2015), so this is a design property, not a proven guarantee for this construction (see Limitations). On every boot, the reconstructed response deterministically derives an elliptic-curve (P-256) key pair via HKDF-based key derivation (RFC 5869) followed by rejection sampling to a uniform scalar; the device registers its public key with the server once, at enrollment, and the corresponding private key is never transmitted, stored, or otherwise disclosed—it exists only transiently in RAM for the duration of a single boot. *Session identification* (Table 3) is asymmetric on the device side and symmetric on the server side: the device signs a server-issued nonce with its private key (ECDSA), verified against its registered public key; the server proves possession of its own pinned, per-node verifier by hashing it with a device-issued nonce, checked against the value pinned at enrollment. Each side also exchanges a fresh ephemeral public key, bound into the same proofs; the resulting ECDH secret, not the long-term verifier alone, anchors the session key (Table 3), and neither proof nor ephemeral key is reused across sessions. The device's NVS holds only the non-secret stable-bit mask/ECC

data (P_{pub}) and the pinned per-node server verifier V_B; its own private key (sk_A, derived fresh from PUF_A each boot) and its ephemeral key pair exist only transiently in RAM (Section 3).

Server identity: the server is not a good physical candidate for PUF-based identity (no comparable unclonable substrate is assumed for the RPi4), so its identity parameter is a conventionally-provisioned verifier V_B, generated fresh per enrolled device and pinned on that device as its server-authentication trust anchor. V_B is deliberately not shared across the fleet: recovering one device's pinned V_B lets an attacker impersonate the server only to that single device, and (Table 3; Limitations) no longer by itself discloses that device's past or future P3 traffic. Server authentication is symmetric: possession of V_B is exactly what the server must prove per session, via the same nonce-bound hash structure that gives the device's proof its freshness guarantee; its confidentiality on the device side is bounded by physical security (Information Disclosure, Table 6; Limitations). Device authentication, by contrast, is asymmetric: the server stores only each device's public key, so a full server compromise discloses no value sufficient to impersonate any device (Section 5.6). The two entities' parameters are thus deliberately asymmetric, bounding the single point of failure a fully-symmetric scheme would place on the server.

Asymmetric scheme (capable devices): Entities generate public/private key pairs, exchange public keys with identifiers, and store the peer's public key for subsequent authentication using ISO/IEC 9798-3-style signature-based mutual authentication (ISO/IEC, 2019) (Table 4).

Table 2: Root of Trust process: PUF-seeded device identity establishment — Phase 1 (enrollment and registration, performed once, over an isolated/physical provisioning channel). PUF_A denotes the device's PUF response, reconstructed from RTC FAST SRAM via `esp32_puflib`; $\texttt{mask}_A/\texttt{ecc}_A$ are the non-secret stable-bit mask and error-correction helper data. (sk_A, pk_A) is the device's P-256 key pair, derived deterministically from PUF_A together with $H_{mask/ecc}$, a hash of the stable-bit-mask/ECC helper data (Table 3, Limitations); only pk_A (public) is ever registered or stored off-device. V_B is a conventionally-provisioned, per-node server verifier (Section 4, "Server identity"), not PUF-derived and not shared across nodes.

Device	**Message**	**Server**
$\{ID_A\}$ enroll_puf() $\rightarrow \{\text{mask}_A,\ \text{ecc}_A\}$ (stored, public) get_puf_response() $\rightarrow \text{PUF}_A$; $(sk_A,\ pk_A) \leftarrow$ HKDF-derive-keypair $(\text{PUF}_A,\ H_{mask/ecc})$		$\{ID_B,\ URL/IP_B,\ Port_B\}$ (fleet-wide, non-secret)
	$\{ID_A,\ pk_A\} \rightarrow$ $\leftarrow \{ID_B,\ URL/IP_B,\ Port_B,\ V_B\}$	generate fresh per-node V_B; register $\{ID_A,\ pk_A,\ V_B\}$
pin (store): $\{ID_B,\ V_B\}$ as server reference		store $\{ID_A,\ pk_A\}$ as device reference (pk_A public, not a secret)

Table 3: Root of Trust process: PUF-seeded device identity establishment — Phase 2 (per-boot reconstruction and mutual challenge; continues Table 2). Device authentication is asymmetric (sk_A never transmitted, verified via pk_A alone); server authentication is a nonce-bound hash-reconstruction proof over V_B. Each side also exchanges a fresh, session-only ephemeral public key (pk_{eph_A}/pk_{eph_B}), bound into both proofs via the transcript hash $T = \text{SHA-256}(\tau \,\|\, ID_A \,\|\, ID_B \,\|\, N_B \,\|\, N_A \,\|\, pk_{eph_A} \,\|\, pk_{eph_B} \,\|\, R_B)$ (τ a fixed domain-separation tag) against substitution or replay. Key_{ab} derives from the resulting ECDH secret Z together with V_B, giving forward secrecy (see main text, Limitations). $H_{mask/ecc} = \text{SHA-256}(\text{mask}_A \,\|\, \text{ecc}_A \,\|\, \text{mask}_A^{sleep} \,\|\, \text{ecc}_A^{sleep})$ is a hash of all four stable-bit-mask/ECC helper-data blobs (both the RTC and the deep-sleep-retry pair, Limitations), bound into sk_A's derivation so that tampering with any of them changes the derived key unpredictably rather than leaking information through the reconstruction's error count. PUF_A/sk_A and eph_A are all freshly (re)generated each boot; nothing here persists between Phase 1 and Phase 2.

Device	**Message**	**Server**
generate nonce N_A	$\{ID_A,\ N_A\} \rightarrow$	
	$\leftarrow$	generate nonce N_B
get_puf_response() $\rightarrow \text{PUF}_A$; re-derive $(sk_A,\ pk_A)$ as in Table 2; generate ephemeral $(sk_{eph_A},\ pk_{eph_A})$	$\{ID_B,\ N_B,\ pk_{eph_B},\ R_B\}$, $R_B = \text{SHA-256}(V_B \,\|\, N_A \,\|\, pk_{eph_B})$	generate ephemeral $(sk_{eph_B},\ pk_{eph_B})$; compute R_B from stored V_B
verify $R_B \stackrel{?}{=} \text{SHA-256}(V_B \,\|\, N_A \,\|\, pk_{eph_B})$ using pinned V_B (server authentication, binds pk_{eph_B}); $R_A \leftarrow$ ECDSA-sign$(sk_A,\ T)$	$\{pk_{eph_A},\ R_A\} \rightarrow$	ECDSA-verify$(pk_A,\ T,\ R_A)$ using registered pk_A (device identification, binds the full transcript T)
	$\leftarrow$ {session_ack} (or rejection if verification failed)	issue `session_ack` only if R_A verifies
$Z \leftarrow$ ECDH$(sk_{eph_A},\ pk_{eph_B})$; SHA-256$(Z \,\|\, V_B \,\|\, N_A \,\|\, N_B) \rightarrow Key_{ab}$		$Z \leftarrow$ ECDH$(sk_{eph_B},\ pk_{eph_A})$; SHA-256$(Z \,\|\, V_B \,\|\, N_A \,\|\, N_B) \rightarrow Key_{ab}$

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Table 3 continued from previous page

Device	Message	Server
Key_{ab} held only in RAM for session duration; sk_A discarded with PUF_A, sk_{eph_A} and Z discarded, at end of boot/session		Key_{ab} held in RAM; sk_{eph_B} and Z discarded after use; pk_A reference retained in server storage (public, not sensitive)

Implementation note: the implementation source (`smg_puf_rot_node/`, device side; `smg_primary_server/rot_challenge_service.py`, server side; companion code repository, Materials Availability) uses `shs_b/V_B` for V_B, `challenge_a/challenge_b` for N_A/N_B, `eph_pk_a/eph_pk_b` for pk_{eph_A}/pk_{eph_B}, `server_proof/device_proof` for R_B/R_A (`device_proof` carries a DER-encoded ECDSA signature, not a fixed-length hash), and a fixed ASCII literal (`DEVICE_PROOF_DOMAIN_TAG`, identical on both sides) for τ; the four-message sequence and the Key_{ab} derivation above match the implementation exactly (the device's local steps are listed in logical order; the implementation reconstructs the response, derives (sk_A, pk_A), and generates its ephemeral key pair before drawing N_A and sending the first message), including concatenation order, the uncompressed-point encoding of both ephemeral public keys, and the `utils.Prehashed` convention the server uses to verify a signature computed over an already-hashed message.

4.7. Threat Coverage by Design: STRIDE Analysis

We analyze how the methodology's stages address the STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege) threat categories (Shostack, 2014) for centralized IoT ecosystems. Unlike framework-specific threat models tied to particular technologies, this analysis maps threats to methodology stages and technology-agnostic mechanism categories, ensuring applicability across implementations.

Tables 5 and 6 map each STRIDE threat to the methodology stage(s) providing countermeasures, the mechanism category mandated by those stages, and the instantiation in the SMG case study. The adversary $\mathcal{A}$ is assumed to have network access, commodity computational resources, and knowledge

Table 4: Root of Trust process: asymmetric key establishment.

Device	**Message**	**Server**
$\{ID_A,\ MAC_A,$ $PubKey_A,\ PrivKey_A\}$		$\{ID_B,\ URL/IP_B,$ $Port_B,\ PubKey_B,$ $PrivKey_B\}$
	$\{ID_A,\ MAC_A,\ PubKey_A\}$ $\rightarrow$ $\leftarrow \{ID_B,\ URL/IP_B,$ $Port_B,\ PubKey_B\}$	
secure store: $\{PubKey_B,$ $ID_B,\ URL/IP_B,\ Port_B\}$		secure store: $\{PubKey_A,\ ID_A,\ MAC_A\}$

of the system architecture (Kerckhoffs's principle); goals include unauthorized data access, service disruption, entity impersonation, and credential exfiltration.

Table 5: STRIDE threat model: mapping to methodology stages and SMG instantiation — Part 1 (Spoofing, Tampering, Repudiation).

Threat	Attack Vector	Methodology Stage(s)	Mechanism Category	SMG Instantiation	Residual Risk
Spoofing	Entity impersonation; forged credentials	2.2.2 (auth mech.), 2.2.3 (RoT), 6.1.3 (RoT execution)	Cryptographic mutual authentication using entity-specific, PUF-derived identity commitments	PUF-derived identity (`esp32_puflib`) + asymmetric device proof/symmetric server proof; device's private key never leaves RAM; handshake before any data accepted; $N = 30$ legitimate handshakes all succeeded (Section 6.3), but FT-06's cross-chip cloning test (11/11 rejected, across four firmware generations), not a successful login, is the direct impersonation-resistance evidence	Low (requires physical device compromise)

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Table 5 continued from previous page

Threat	Attack Vector	Methodology Stage(s)	Mechanism Category	SMG Instantiation	Residual Risk
Tampering	Data modification in transit; sensor reading manipulation	2.2.4 (secure comms), 5.2.1 (secure dev.)	Authenticated encryption with integrity binding per message; random nonces	AES-256-GCM per message; random 12-byte nonce; GCM tag detects single-message modification	Low per-message (GCM tag); Medium for re-ordering/replay (no message-sequence chain, see Limitations)
Repudiation	Denial of sent data; dispute of control commands	2.2.2 (session audit), 3.1 (data service), 7.1 (monitoring)	Session-bound data exchange with server-side session logging	Server logs each session under `node_id` and timestamp (Stage 7.1); evidentiary, not an independently-verifiable chain (trade-off, see Limitations)	Medium (logs can be disputed without independent cryptographic chain verification)

Table 6: STRIDE threat model (continued from Table 5) — Part 2 (Information Disclosure, Denial of Service, Elevation of Privilege).

Threat	Attack Vector	Methodology Stage(s)	Mechanism Category	SMG Instantiation	Residual Risk
Information Disclosure	Key extraction; credential exfiltration; traffic interception	1.4.2–1.4.3 (sovereignty), 2.2.1 (identity params), 5.2.1 (secure dev.)	Data sovereignty policy; no persistent device-own secret to disclose; application-layer encryption	Node identity reconstructed at boot from the PUF, never persisted; NVS holds only non-secret values for the node's own identity (helper data, `id_a`, `id_b`, V_B, WiFi credentials), no private key or raw PUF response; AES-256-GCM keyed from V_B plus a discarded per-session ECDH secret (forward secrecy)	Node identity: Low (server stores only pk_A). Server impersonation from a captured unit: Medium for live impersonation (V_B alone suffices to complete a real-time handshake); Low for decrypting a recorded session, since V_B alone no longer determines Key_{ab} (see Limitations)

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Table 6 continued from previous page

Threat	Attack Vector	Methodology Stage(s)	Mechanism Category	SMG Instantiation	Residual Risk
Denial of Service	Authentication flooding; bus load disruption; node isolation	7.1 (monitoring), 7.2.1 (anomaly detect.), 2.1.8 (hierarchy)	Anomaly-based trust scoring; degraded-mode operation; entity isolation policy	Trust score decay on repeated failures; UNTRUSTED nodes isolated (TCP dropped); last PWM duty maintained on timeout (original server modules; not yet ported to the RoT server, see Limitations)	Medium (physical-layer DoS outside software scope)
Elevation of Privilege	Node masquerading as admin; unauthorized config access	2.1.8 (entity hierarchy), 3.2.1 (access control), 2.2.2 (auth)	Entity hierarchy with entity type determined by identification channel, not a field inside a shared message	Nodes authenticate exclusively through the P1–P3 RoT protocol; users/administrators use a separate credential-based API; P3 data-channel access is bound to the P2 session key, so a node session cannot invoke user-/admin-scoped endpoints	Low (no message format allows claiming a different entity type)

5. Case Study: DC Smart Microgrid IoT Ecosystem

5.1. Case Study Design

This case study follows a **confirmatory design** (Yin, 2014): the methodology constitutes the theoretical proposition; the SMG implementation constitutes the empirical test. Evidence is collected from four sources following Yin's principle of multiple evidence sources (Table 7).

Table 7: Data collection protocol: evidence sources.

Source	**Evidence Type**	**Content**
Design documents	Documentary	Stage outputs: policies, architecture diagrams, identity tables, service specifications
Firmware artifacts	Physical artifact	ESP32 C/ESP-IDF modules: identity (PUF enrollment/reconstruction, `esp32_puflib`), Root of Trust handshake, secure communication (AES-256-GCM)
Execution logs	Archival records	Authentication logs, RoT deployment logs, functional test reports
Measurements	Quantitative	Identity provisioning, handshake latency, OWASP coverage

5.2. Research Questions

5.3. Case Context

Application domain: The SMG manages distributed DC electrical energy generation, storage, and consumption across two co-located nodes in a controlled laboratory environment. The growing integration of renewable energy sources (solar panels, batteries) into local grids requires intelligent, automated management to balance generation and demand in real time without relying on centralized national grid infrastructure. This application is classified as critical infrastructure (Methodology Stage 1, Process 1.4.1), requiring the highest cybersecurity tier.

Physical architecture: The SMG consists of two secondary control agent nodes and one primary agent (Raspberry Pi 4B). Each secondary node connects a renewable generation element to a shared 12 V DC bus with local

Table 8: Case study research questions.

ID	Question	Addressed by
CSQ1	Does the methodology provide sufficient guidance to architect a complete, secure IoT ecosystem for a real industrial application?	Stage-by-stage traceability; completeness of outputs; functional test results
CSQ2	Does the parameterized identity model enable unambiguous security mechanism definition across heterogeneous entity types?	Identity parameter table; authentication implementation; coverage across 4 entity types
CSQ3	Does the Root of Trust Process establish verifiable secure communication between ecosystem entities?	RoT execution evidence; handshake latency measurements; mutual challenge-response verification
CSQ4	What is the methodology's coverage of known IoT security threats relative to established standards?	OWASP IoT Top 10 mapping; security mechanism traceability

energy storage (Table 9). The primary agent runs on a Raspberry Pi 4B connected to the DC bus as a pure load via a DC/AC inverter (12 V DC → 120 V AC) followed by a USB-C AC/DC adapter.

Table 9: Secondary node hardware components.

Component	**Role**	**Specification**
ESP32 NodeMCU	IoT node controller	Dual-core, rated up to 240 MHz¶, 520 KB SRAM, 4 MB Flash, built-in WiFi; powered by local battery via DC-DC buck (12V→5V)
Epcom AGM VRLA	Local energy storage + ESP32 supply	PL-18-12, 12 V, 18 Ah; also supplies ESP32 via buck converter
IBFIA DC-DC SSR ×3	Energy flow control	3–32 VDC control, 5–60 VDC load, 40 A; driven by ESP32 LEDC on GPIO27/26/25 at 1 kHz, 10-bit
Schneider MCB ×3	Overcurrent protection	2-pole, one per SSR channel
Solar charge controller	Energy flow control	Commercial MPPT controller (handles droop control and battery-charging voltage regulation autonomously)
ACS712-05B ×2	Current sensing	Hall-effect, 185 mV/A, 16-sample median oversampling
Resistive divider ×2	Voltage sensing	30 kΩ/7.5 kΩ (5:1 ratio); reads up to 16.5 V on 3.3 V ADC

¶Rated maximum; the current firmware runs the CPU at 160 MHz (bench boot logs, Section 6.3), the configuration the reported latencies reflect.

Digital architecture: The ecosystem specifies a centralized communication topology (Stage 2.1.4) with fog processing (Stage 2.1.7): a 9-rule fuzzy EMS per ESP32 and a 27-rule security-aware EMS on the RPi4, neither yet implemented (Limitations). The entity hierarchy is: IoT Nodes (lowest privilege) < Users < Administrators. Development target: TRL 3 (proof of concept with physical elements).

5.4. Stage-by-Stage Traceability

Tables 10 and 11 document the application of each methodology stage to the SMG, showing the decision made at each stage and the measurable artifact produced. This traceability table constitutes the primary evidence for CSQ1.

Table 10: Stage-by-stage traceability: methodology process → SMG decision → artifact — Part 1 (Stages 1–3).

Stage	Methodology Process	Decision Made in SMG	Artifact
1 Techno-logical Paradigm	1.1 Establish problem	Manage distributed renewable energy nodes securely in a 12 V DC microgrid	Problem specification
	1.1.2 Entities involved	5 entity types: IoT nodes, server, users, admins, DC bus	Entity list
	1.4.1 Criticality	Classified as critical infrastructure (electrical energy)	High-security cybersecurity policy
	1.4.2 Technological sovereignty	Self-hosted; no third-party cloud; ESP32 + local RPi4 server	Architecture constraint record
	Of the seven conceptual-framework element types (Section 3), the SMG instantiates five: storage/communication infrastructure are subsumed into the server, and IoT services into the device/server software. Of the five entity types, four (nodes, server, users, admins) carry digital identity (Table 12); the DC bus is tracked via P3 sensor-data integrity instead.		
2 Digital Ecosystem	2.1.4 Communication topology	Centralized: all nodes report to primary agent	Architecture diagram
	2.1.7 Processing topology	Fog: local EMS on ESP32; global coordination on RPi4	Processing topology record
	2.2.1 Identity definition	4 entity types, 14 identity parameters total (non-uniform per type; Table 12)	Identity parameter table

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Table 10 continued from previous page

Stage	Methodology Process	Decision Made in SMG	Artifact
	2.2.2 Authentication mechanism	Asymmetric (ECDSA) PUF-seeded device proof + symmetric, conventionally-provisioned per-node server proof; AES-256-GCM session encryption	Authentication diagram
	2.2.3 Root of Trust process	4-phase RoT designed for node–server pairs	RoT execution log
3 IoT Services	3.1 Service definition	4 services specified: data ingestion, EMS coordination, auth API, monitoring; only RoT-based node authentication is currently implemented (Limitations)	Service specification
	3.2.1 Access control	Entity type determined by identification channel (Section 5.6), not a token field; a separate token-based admin/user API is specified but not yet ported (Limitations)	Access control matrix

Table 11: Stage-by-stage traceability (continued from Table 10) — Part 2 (Stages 4–7).

Stage	Methodology Process	Decision Made in SMG	Artifact
4 Tech. Modular Integration	4.1.1 TRL planning	TRL 3 target for ITER 1; lab. proof of concept	TRL milestone record

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Table 11 continued from previous page

Stage	Methodology Process	Decision Made in SMG	Artifact
	4.1.2 Platform selection	ESP32 + C/ESP-IDF: WiFi, SHA-256 HW, NVS, `esp32_puflib` as native component	Technology selection rationale
	4.1.3 Communication	WiFi IEEE 802.11 b/g/n + TCP; application-layer mutual-authentication protocol (`rot_session.c`/ `rot_challenge_ service.py`)	Protocol selection record
	4.2 Vulnerability mitigation	ESP-IDF v5.4.2 used for every reported build (recorded in the build logs and stated as a requirement in the code repository's README); a per-session random nonce for every P2/P3 message (no nonce reuse)	Risk register
5 Applications & Devices	5.1.1 Server	`rot_challenge_service.py` on RPi4 (raw TCP/JSON); REST API/time-series DB/dashboard specified but not yet ported (Limitations)	Server deployment
	5.1.2 Devices	ESP32 C/ESP-IDF firmware: identity (PUF), Root of Trust, secure-communication modules; sensor/EMS not yet implemented (see Limitations)	Firmware artifact

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Table 11 continued from previous page

Stage	Methodology Process	Decision Made in SMG	Artifact
	5.2.1 Secure development	NVS stores only non-secret mask/ECC helper data, plus `id_b`/V_B/WiFi SSID-password (no device-private secret persisted); `esp_fill_random()` nonces; no credential value in firmware source or binary	Code review record
6 Testing & Deployment	6.1.3 RoT execution	Validated end-to-end on 2/2 nodes: NODE_01 ($N = 30$, 30/30, detailed bench Table 17) and NODE_02 ($N = 30$, 30/30, confirming cross-chip reliability; Section 6.3)	PUF bench (Section 6.3)
	6.2.1 Functional testing	Boot, PUF reconstruction, mutual authentication (P2), encrypted exchange (P3) all pass across $N = 30$ trials; sensor/EMS tests not yet applicable	PUF bench (Section 6.3)
	6.2.2 Tamper verification	NVS tampering confirmed as auth. failure (three tests, FT-05; full pen. testing deferred to ITER 2)	Serial boot and server logs (`hash_coverage_tamper_test/`)
7 Operation & Mgt.	7.1.1 Monitoring	Authentication status and session history from server-side logs; V/I/P/PWM telemetry is specified but not yet implemented (no sensor module, Limitations)	Dashboard artifact
	7.2.2 Update elements	OTA update policy defined with authentication requirement (ITER 2 target)	Update policy

5.5. Identity Parameter Definition (Stage 2.2.1)

Applying Equations (1)–(3) to the SMG entities produces a 14-parameter identity model (Table 12). The public/private/authentication parameter separation directly determined the authentication mechanism (PUF-derived commitment using the private, non-persistent parameter) and the secure storage requirement for the IoT Node: none, beyond the non-secret mask/ECC helper data already classified as public.

Table 12: Identity parameter table: Stage 2.2.1 output for the SMG. $\texttt{mask}_A/\texttt{ecc}_A$: non-secret stable-bit mask and error-correction helper data generated at enrollment by `esp32_puflib`, stored in ESP32 NVS. PUF response†: the device's physically-derived private identity parameter, reconstructed from $\texttt{mask}_A/\texttt{ecc}_A$ on every boot and never persisted (see note below). $V_B^\ddagger$: the server's per-node authentication verifier (see note below). `id_a`§: the device's wire identifier, now per-device NVS-provisionable (see note below). Following Eqs. (2)–(3), $n = n_X^E$ is the public-parameter count, $k = k_X^E$ the private-parameter count, and Total $= n_X^E + k_X^E$ the total identity-parameter count.

Entity Type	**Class**	P_{pub}	P_{priv}	P_{auth}	n	k	**Total**
IoT Node (ESP32)	Device	`id_a`§, $\texttt{mask}_A$, $\texttt{ecc}_A$	PUF response†	PUF response†	3	1	4
IoT Server (RPi4)	Server	`url/ip`, `port`, `id_b`	$V_B^\ddagger$	$V_B^\ddagger$	3	1	4
User	User	`username`	`user_uuid`, `password`	`user_uuid`, `password`	1	2	3
Administrator	Admin	`username`	`admin_uuid`, `password`	`admin_uuid`, `password`	1	2	3
Total		**8**	**6**	**6**			**14**

The entity-specific separation of identity parameters produced a direct, unambiguous mapping to implementation artifacts: $\texttt{mask}_A$ and $\texttt{ecc}_A$ correspond to distinct NVS keys written at enrollment time; no parameter redefinition was required at the specification level for those two. `id_a` now follows the same path when provisioned: it defaults to a compile-time placeholder shared by every unit, but a per-device NVS value overrides it when provisioned (Table 12, note §; see Limitations for the fix history). In the present case study, P_{auth} and P_{priv} coincide for every entity type (all private parameters participate in authentication, as shown in Table 12); this is a case-study-specific coincidence, not a general property of the model—Campos-Sanchez et al. (2026a) define only the broader relation $P_{auth} \subset I$

(Figure 3), so future case studies may include a private parameter not used for authentication (e.g., a long-term audit key), in which case $P_{auth} \subsetneq P_{priv}$ rather than $P_{auth} = P_{priv}$.

[†]**Note on the PUF response as a non-persistent private parameter:** the formal identity model $I_X^E = P_{X\,pub}^E \cup P_{X\,priv}^E$ (Section 4) classifies a parameter as private or public by its confidentiality role, not by whether it is stored. The IoT Node's PUF response satisfies P_{priv}'s role—it is never disclosed and underlies the device's authentication proof—while never existing as data at rest: it is regenerated from $\texttt{mask}_A/\texttt{ecc}_A$ (both P_{pub}) on every boot. This is the concrete instance of the Phase 2 (secure storage) trivialization noted in Section 3.

[‡]**Note on V_B as the server's private/authentication parameter:** the server's row describes its own proof of identity to a device, made by demonstrating knowledge of V_B (Table 3) — the server-side analogue of the Node's PUF response. `id_b` (fleet-wide, non-secret) sits under P_{pub} with `url/ip` and `port`. The device separately pins its own copy of V_B as a trust anchor (Section 4, "Server identity"); this is a stored reference to *another* entity's parameter, not an additional Node parameter, so it is not double-counted.

[§]**Note on `id_a`:** defaults to a compile-time placeholder shared by every unit, overridden by a per-device NVS value when provisioned (`tools/provision_server_ref.py -id-a`); it identifies the device on the wire and, as a public value, is bound into the signed P2 transcript (Table 3) alongside the nonces and ephemeral keys, but it is not itself a secret or the cryptographic primitive the proof rests on – that is the PUF-derived key pair (see Limitations for the fix history).

5.6. Security Mechanism Analysis: STRIDE–to–Protocol Mapping

The methodology-level STRIDE analysis (Section 4.7) identifies mechanism categories that address each threat category at the framework level, independent of implementation technology. This subsection demonstrates how the SMG's Stage 4 technology selection instantiates those mechanisms through three authentication and data-exchange protocol phases (P1–P3), establishing a complete traceability chain from threat identification to deployed countermeasure.

Scope note: Phases P1–P3 constitute an *implementation-specific* Stage 4 technology choice for the SMG (see Introduction scope note). Any protocol providing equivalent mechanism categories—pre-deployment

identity isolation (P1), nonce-bound mutual identification without key transmission (P2), and per-message authenticated encryption (P3)—would satisfy the same STRIDE and OWASP properties against the same methodology requirements. This design intentionally omits a cryptographic session-chain/audit mechanism; Tables 5/6 and the Limitations (Section 7) record the resulting Repudiation/I8 trade-off explicitly.

Table 13 describes each phase's security role, mechanism, and STRIDE/OWASP coverage. Table 14 summarizes the STRIDE coverage matrix across phases.

Table 13: SMG authentication protocol phases: security role, mechanism, and coverage mapping (P1–P3). STRIDE: S = Spoofing, T = Tampering, R = Repudiation, ID = Information Disclosure, DoS = Denial of Service, EoP = Elevation of Privilege.

Ph.	**Name**	**Security Role**	**Mechanism**	**STRIDE**	**OWASP**
P1	Enrollment	Identity establishment before any network exposure	`enroll_puf()` (`esp32_puflib` (Stanicek, 2022)): 20 RTC-FAST-SRAM measurements → stable-bit mask + 8×-repetition-code ECC helper data (both non-secret), written to ESP32 NVS via USB serial on isolated workstation; WiFi SSID/password are provisioned into the same NVS image out-of-band (`tools/provision_server_ref.py`), never compiled into firmware source (the same mechanism optionally pins a per-device `id_a`, see Limitations); device public key pk_A registered with the server, which generates and pins a fresh per-node verifier V_B in return; no raw PUF response or private key is ever written to flash (a library defect that briefly violated this was found and fixed during this work, see Limitations)	S, ID, EoP	I1, I9

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Table 13 continued from previous page

Ph.	Name	Security Role	Mechanism	STRIDE	OWASP
P2	Root of Trust	Per-boot response reconstruction and nonce-bound mutual identification without key transmission; ephemeral ECDH session key (forward secrecy)	`get_puf_response()` reconstructs PUF_A (see Section 6.3 for the measured byte length, which varies slightly by enrollment); (sk_A, pk_A) re-derived as in Table 2, now also bound to a hash of the stable-bit mask/ECC helper data (Limitations); device proves possession of sk_A via ECDSA signature over a server-issued nonce and both ephemeral public keys; server proves possession of this node's V_B via a nonce-bound hash (Table 3); $Key_{ab} \leftarrow$ SHA-256$(Z \,\|\, V_B \,\|\, N_A \,\|\, N_B)$, Z a discarded ECDH secret; none of the raw response, sk_A, V_B, or Z is ever transmitted; validated end-to-end on real hardware (Section 6.3)	S, ID, EoP; T, DoS (partial)	I1, I3
P3	Data Exchange	Per-message authenticated encryption	$c_i \leftarrow$ AES-256-GCM$(Key_{ab}, \text{nonce}_i, m_i)$ via ESP-IDF `mbedtls/esp_aes`; a fresh random 12-byte nonce per message; the GCM authentication tag detects any single-message modification	T, ID; R (partial)	I2, I3, I6, I7

Spoofing is countered at two layers: P1 ties valid identity to physical access to the specific enrolled chip rather than a copyable stored secret; P2's nonce-bound challenge requires fresh, per-session proof from each side—an ECDSA signature from the device, a hash-reconstruction proof over V_B from the

Table 14: STRIDE coverage matrix across protocol phases. ✓ = primary countermeasure (first line of defense); ~ = secondary/partial contribution; — = not applicable.

STRIDE Threat	**P1**	**P2**	**P3**
Spoofing	✓	✓	—
Tampering	—	~	✓
Repudiation	—	—	~
Info. Disclosure	✓	✓	✓
Denial of Service	—	~	—
Elev. of Privilege	✓	✓	—

server—without revealing the physical response, sk_A, or V_B itself, confirmed bidirectionally before any session key is derived.

Tampering is detected at message creation time by the AES-256-GCM authentication tag on every message (P3); P2's mutual identification additionally binds the session key to both parties' proven identities. The scheme has no session-level hash chain, so retrospective re-verification of message ordering is unavailable (downgraded to ~ in Table 14; see Limitations).

Repudiation coverage is evidentiary rather than cryptographic: P3 messages are authenticated in transit and the server logs each session (`node_id`, timestamp, Stage 7.1), but no independently-verifiable chain binds messages to a session without trusting the server's logs, a deliberate trade-off for a smaller on-device footprint (see Limitations).

Information Disclosure is addressed at the source (P1: no key ever exists in storage, only non-secret helper data), during derivation (P2: the physical response is reconstructed transiently in RAM and never transmitted; neither sk_A nor V_B is ever sent, only proofs over them), and in transit (P3: per-message AES-256-GCM under a key that also depends on a discarded per-session ECDH secret, with random nonces).

Denial of Service is partially addressed at P2: the server's first reply (R_B) costs one nonce, one ephemeral P-256 key-pair generation, and one SHA-256 hash — modest, but incurred for any requester presenting a registered `id_a`, which defaults to a shared, publicly visible placeholder unless a node has been given a distinct per-device value (Limitations) — yet no further session state is committed—and no `session_ack` issued—until the device's ECDSA proof

R_A verifies, imposing an asymmetric cost on an attacker holding a session open. Volumetric and application-layer flooding beyond the identification gate remain outside P1–P3's scope; Stage 7 trust-based isolation (Table 6) is the mandated mechanism, implemented in the SMG's original server modules but not yet ported to the current RoT server (see Limitations).

Elevation of Privilege is prevented at P1, since entity type is determined by which identification channel a party uses, not a field inside a shared message: nodes authenticate exclusively through the P1–P3 RoT protocol, while users and administrators use a separate, credential-based API (Table 12); no message in Tables 2/3 carries an entity-type field. P3 API access is bound to the P2 session key, so a node session cannot invoke user- or admin-scoped endpoints.

6. Empirical Evaluation

6.1. CSQ1: Architecture Guidance Sufficiency

Traceability completeness: The stage-by-stage traceability tables (Tables 10 and 11) indicate that all 7 stages produced documented, verifiable artifacts—23 distinct process-to-artifact rows across the four phases. No stage required design decisions beyond what the methodology's processes prescribe.

Functional validation: Table 15 reports the functional test results, drawn from the $N = 30$ real hard-reset bench trial (Section 6.3) and the dedicated tamper-verification test.

Table 15: Functional test results (Stage 6.2.1). FT-01–FT-04 pass rates are over the $N = 30$ hard-reset bench trial (Section 6.3); FT-05 is a dedicated negative test (one original run plus two archived runs on the current firmware); FT-06 is a cross-chip cloning negative test repeated 11 times across four firmware generations (Limitations). Sensor/EMS functional tests are not yet applicable, since those modules are not implemented in the current firmware (see Limitations).

ID	Description	Result
FT-01	Boot: PUF identity reconstructed from NVS helper data (mask/ECC); no raw key or PUF response in NVS	Pass (30/30)

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Table 15 continued from previous page

ID	Description	Result
FT-02	WiFi connect: associates to the configured access point and obtains an IP address	Pass (30/30)
FT-03	P2 Root of Trust: asymmetric device proof (ECDSA) + symmetric server proof both verify, over an ephemeral ECDH exchange; mutual authentication established	Pass (30/30)
FT-04	P3 session: AES-256-GCM encrypted demo message sent by the device and decrypted server-side	Pass (30/30)
FT-05	Tamper verification: deliberate NVS parameter corruption produces authentication failure. One original targeted test (log not archived), repeated on the current firmware as two archived single-bit-flip tests, one on each retry-path helper-data blob (Section 6.3)	Pass (3/3)
FT-06	Cross-chip cloning: SMG_NODE_01's entire NVS (PUF helper data, server reference) copied onto SMG_NODE_02's physical chip; server rejects the resulting device proof (signature invalid). Repeated 11 times across four firmware generations (1 original, 4 error-rate-relaxed, 3 first-generation HDM binding, 3 extended HDM binding), all rejected (Limitations)	Pass (11/11)
	Overall: 6/6 tests passed (4/4 positive tests FT-01–FT-04; 2/2 negative tests FT-05, FT-06)	**100%**

Iterative development confirmation: At least seven maintenance moments were identified and resolved during real-hardware bench validation and subsequent review passes (Section 6.3, Limitations): an enrollment-state-confusion bug causing an unbounded re-enrollment loop; the session-key forward-secrecy fix (ephemeral ECDH); moving the WiFi credential from a firmware constant to NVS provisioning; erasing the `PUF_FREQUENCY` blob left in NVS after enrollment; relaxing the miscalibrated local reliability gate; the helper-data-manipulation mitigation (including extending its scope after an initial gap was found); and the per-device `id_a` provisioning fix that enabled registering a second node. Each was a code-level revision triggered by testing

or review, not an architectural redesign. This confirms the methodology's M2 maintenance pathway operates as specified at TRL 3 (see Discussion for the M1/M2 classification).

Answer to CSQ1: The methodology provided sufficient guidance to specify the complete IoT ecosystem and to implement and operate its identity, Root of Trust, and secure-communication layers on real hardware; the sensor/EMS and remaining server-side modules are specified but not yet implemented (Limitations). All 7 stages produced documented outputs with no external guidance required.

6.2. CSQ2: Identity Model Unambiguity

Applying the identity model equations to 4 entity types produced 14 distinct parameters with zero overlap between entity classes (Table 12). The public/private/authentication tripartite structure directly determined: (a) what is public non-secret helper data stored in NVS ($\texttt{mask}_A$, $\texttt{ecc}_A$), (b) what is never persisted and is instead reconstructed on every boot (the PUF response), and (c) what constitutes the device's authentication proof (an ECDSA signature over the P2 transcript, produced with the PUF-derived private key; `id_a` identifies the device on the wire and is bound into that signed transcript, but is not itself the secret the proof rests on). Table 16 summarizes these identity model metrics; both of the SMG's 2 nodes have now been provisioned and validated under this model (Section 6.3).

Table 16: CSQ2 identity model metrics.

Metric	**Value**
Entity types defined	4
Total identity parameters	14
Public parameters (P_{pub})	8
Private parameters (P_{priv})	6
Authentication parameters (P_{auth})	6
Nodes provisioned via RoT	2/2
Parameter collisions (overlapping definitions)	0
Additional design decisions required at Stage 5	0

Answer to CSQ2: The identity model is sufficient and unambiguous. All 4 entity types received complete, non-overlapping identity definitions that translated directly to implementation artifacts without modification.

6.3. CSQ3: Root of Trust Verifiability

The RoT process was executed and bench-validated end-to-end on real hardware for SMG_NODE_01 (ESP32-D0WD-V3 rev 3.1), connected over the real `SMG_Primary` WiFi access point to the real RPi4 primary agent (`rpi4-server`) running the server-side counterpart `rot_challenge_service.py`. **Both of the SMG's nodes are now validated: SMG_NODE_01's detailed $N = 30$ bench is the headline round reported in Table 17 below; SMG_NODE_02 was separately registered and bench-validated (own $N = 30$, 30/30) once a per-device-identifier provisioning gap was closed, confirming the reliability profile generalizes across chips (Limitations).** Post-provisioning, node authentication was verified by deliberate NVS parameter tampering—corrupted parameters produced an authentication failure in every attempt (FT-05: one original test and two archived single-bit-flip tests on the current firmware), confirming that the RoT creates a verifiable security boundary.

Table 17 reports results from $N = 30$ successive hard-reset trials against a single, fixed enrollment (the stable-bit mask and ECC helper data were generated once via `enroll_puf()` and not regenerated between trials); each trial therefore measures per-boot reconstruction-and-handshake reliability, not enrollment-to-enrollment variability, and exercises Phases P2 and P3 of the protocol (Section 5.6) against the real server on every reset, without repeating P1 enrollment.

Table 17: PUF-based RoT real-hardware bench trial: SMG_NODE_01, $N = 30$ successive hard-reset trials against one fixed enrollment. `esp32_puflib`'s own instrumentation reports two per-attempt figures: hamming-weight, a hard gate on the raw measurement (must exceed 48.5%, a randomness sanity check unrelated to any specific enrollment), and raw bit-error rate, reported here for reference only — it does not gate acceptance (Limitations). The latter counts only the bits the ECC decoder corrects within the stable-bit-masked region (422×64 = 27,008 bits for this enrollment), but divides by the full 32768-bit measured region rather than the masked subset, so it understates the error rate over the stable bits actually used; the per-stable-bit figure is proportionally higher by a factor of $32768/27008 \approx 1.21$. The server's ECDSA verification, not either local figure, is what establishes identity.

Metric	**Value** ($N = 30$)	**Notes**
End-to-end success rate	30/30 (100%; two-sided 95% Clopper–Pearson interval 88.4–100%)	Mutual identification (P2, with ephemeral ECDH) + AES-256-GCM demo message exchange (P3), confirmed on both device and server logs; repeated trials on one device/enrollment, not independent samples (see discussion below)
Reconstruction attempts per trial	1/1 in every trial (30/30 succeeded on attempt 1)	No retry triggered; the library's two-tier retry mechanism remains available as a fallback but was not exercised in this round (see discussion below)
Attempt-1 raw bit-error rate	0.397 ± 0.052 % (range 0.308–0.485 %)	Diagnostic only (see discussion below); above the library's original 0.15% figure in every trial, same as every prior round, yet every reconstruction was accepted on the first attempt and verified by the server

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Table 17 continued from previous page

Metric	**Value** ($N = 30$)	**Notes**
Handshake latency, first exchange (connect → server authenticated)	680.0 ± 9.3 ms	Before the first message: PUF reconstruction (≈70 ms), (sk_A, pk_A) derivation incl. the four-blob HDM hash, ephemeral key generation; then the hello/challenge round trip and the server-proof check over V_B; no ECDSA
Handshake latency, second exchange (server authenticated → device authenticated)	359.7 ± 46.7 ms (median 345)	Device's ECDSA sign plus the server's verify, over the full transcript including both ephemeral keys, plus the response/acknowledgement round trip
Handshake latency, key derivation (device authenticated → session established)	273.7 ± 4.8 ms	ECDH shared-secret computation plus final SHA-256 derivation of Key_{ab}
Handshake latency, total (TCP connect → session established)	median 1300 ms (mean 1313.3 ± 47.6 ms; range 1290–1520)	About 50 ms slower by mean than the previous (first-generation, two-blob) HDM-mitigation round (median 1250 ms, mean 1262.7 ± 47.9 ms), consistent with reading and hashing roughly 7.5 KB more NVS data for the two additional helper-data blobs; about 245 ms slower by mean than the pre-mitigation round (median 1060 ms, mean 1067.7 ± 23.6 ms) (see discussion below)

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Table 17 continued from previous page

Metric	**Value** ($N = 30$)	**Notes**
WiFi association latency	3545.3 ± 20.8 ms (range 3520–3640)	ESP-IDF stack retried the first association attempt in all 30 trials; a WiFi-stack characteristic, not a RoT protocol property
Device key-pair (pk_A) reproducibility	Consistent with a single registered value in 30/30 trials	All 30 trials authenticated against the one server-registered public key; confirmed indirectly, since the raw key material is not logged

Four findings stand out. First and second, two defects were found and fixed during this validation, not pre-existing design properties: a local reliability gate miscalibrated to reject every reconstruction outright (an earlier round under that gate, `bench_results_20261015/`, returned 0/30; this fix did not change the derived key pair), and a helper-data-manipulation oracle in the key derivation whose first mitigation attempt left one of two helper-data pairs unbound (this fix, in both of its passes, does change the derived key pair, requiring re-enrollment each time); both are detailed, with their discovery and fix history, in Limitations. The round reported here (`bench_results_20261016_hdm2/`) reflects both fixes fully applied, after two earlier, now-superseded rounds captured the intermediate states (dataset pointers below). That the second fix's retry-path binding is genuinely load-bearing, not merely read without effect, was confirmed directly: a single bit flipped in either retry-path helper-data blob (leaving the other three blobs and the physical PUF untouched) caused the server to reject the signature, while first-attempt reconstruction — which never reads those blobs — kept reporting a normal error rate; restoring the unmodified image immediately re-established a successful handshake. Third, the end-to-end protocol—PUF reconstruction, nonce-bound mutual identification in both directions, and AES-256-GCM data exchange—succeeded in every trial, confirming a stable identity (and key pair) despite per-attempt measurement noise that now only reaches the server's cryptographic

check. Fourth, the handshake latency splits into three phases, detailed in Table 17: device-side PUF reconstruction and key-pair derivation followed by server authentication, the device's ECDSA sign/verify, and ECDH key derivation; total latency is tightly clustered (median 1300 ms, mean 1313.3 ± 47.6 ms) but about 50 ms slower by mean than the first-generation (two-blob) HDM round (median 1250 ms, mean 1262.7 ± 47.9 ms) and about 245 ms slower by mean than the pre-mitigation round (median 1060 ms, mean 1067.7 ± 23.6 ms); the first phase rose by 44 ms relative to the first-generation HDM round (Table 17), consistent with, though not proof of, the additional NVS reads and hashing the extended binding performs during key-pair derivation, before the first message is sent. A separate, later comparison under a different firmware build (Limitations, per-device `id_a` fix) found a phase-1 difference an order of magnitude larger than this one with no corresponding code change in that phase, so this attribution should be read as plausible, not established. The server log for this round shows 33 "handshake OK" entries (30 reported trials plus 3 incidental verification/restore handshakes) and 5 "FAILED" entries (2 single-bit-flip tamper trials plus 3 FT-06 trials below). The 30 trials are repeated measures on one device/enrollment, not independent samples (Limitations). Raw logs and summaries for all six bench rounds are in the companion artifact repository under `case study SMG/Implementation/smg_puf_rot_node/` (Materials Availability): `bench_results_20261016_hdm2/` is the round reported here; `bench_results_20261015_hdm/` (30/30, first-generation two-blob HDM binding; mean 1262.7 ± 47.9 ms), `bench_results_20261015_fixed/` (30/30, same relaxed error-rate gate but the pre-HDM-mitigation identity; mean 1067.7 ± 23.6 ms), `bench_results_20261015/` (0/30, the miscalibrated-gate round above), `bench_results_20261013/` (30/30, forward-secrecy added but the retry path's error-rate gate not yet enforced; mean 1131 ± 50 ms), and `bench_results_20261005/` (pre-forward-secrecy protocol; mean 558.0 ± 181.2 ms) are all superseded and kept for provenance. All reported standard deviations in this paper are population (not sample) statistics.

Answer to CSQ3: The RoT process established verifiable mutual authentication for SMG_NODE_01 across $N = 30$ successive hard-reset trials (30/30 end-to-end success, Table 17): the device authenticated the server via a nonce-bound hash-reconstruction proof over a per-node verifier, and the server authenticated the device via an ECDSA signature verified against the device's registered public key, with neither the device's private key, the server's per-node verifier, nor the ECDH shared secret ever transmitted. Me-

dian handshake latency was 1300 ms (mean 1313.3±47.6 ms), splitting across device-side PUF reconstruction and key-pair derivation plus server authentication (680.0 ms), the device's ECDSA sign/verify (359.7 ms), and ECDH key derivation (273.7 ms, Table 17). The forward-secrecy fix added the ephemeral key-pair generation (first phase) and the ECDH computation (third phase, which did not exist in the original protocol); the helper-data-manipulation fix added its cost to the first phase, where the key pair is derived (Limitations). SMG_NODE_02 was separately registered and validated under this scheme (own $N = 30$ trial, 30/30, Limitations), confirming the reliability profile is not specific to one physical chip.

6.4. CSQ4: Threat Coverage

Tables 18 and 19 map each OWASP IoT Top 10 item to the methodology stage(s) mandating the countermeasure, the protocol phase(s) that provide the implementation-level evidence (cross-referenced with the phase analysis in Section 5.6 and Table 13), and the specific SMG mechanism.

Table 18: OWASP IoT Top 10 coverage analysis — Part 1 (I1–I5). Stage = methodology process mandating the countermeasure. Phase(s) = protocol phase(s) providing implementation evidence (see Table 13); — = addressed by policy/deployment, not protocol.

#	OWASP IoT Risk	Stage(s)	Phase(s)	Evidence in SMG	Coverage
I1	Weak/ hardcoded passwords	2.2.1	P1, P2	No password or key is used for device identity: reconstructed at boot from a PUF using only non-secret mask/ECC helper data, never a stored password; confirmed working in 30/30 real trials (Section 6.3). The separate WiFi credential is provisioned into NVS out-of-band (`provision_server_ref.py`), not compiled into firmware source or the public binary; it remains one passphrase shared by every unit on the fleet's WPA2-PSK network, which is inherent to WPA2-PSK rather than a firmware defect (see Limitations)	Full
I2	Insecure network services	2.2.4	P2, P3	P2 establishes mutual identification and a session key before any data is accepted; P3 encrypts every message with per-message random 12-byte nonce (AES-256-GCM); no plaintext channel exists	Full

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Table 18 continued from previous page

#	OWASP IoT Risk	Stage(s)	Phase(s)	Evidence in SMG	Coverage
I3	Insecure ecosystem interfaces	3.2.1	P2	P2 nonce-bound mutual identification, both parties verified before any session key is derived; node vs. user/admin access separated by identification channel (STRIDE EoP, Section 5.6); trust scoring exists but not yet ported to the RoT server (Limitations)	Full
I4	Lack of secure update	7.2.2	—	OTA update policy (Stage 7.2.2): admin authentication + firmware signature required; implementation scoped to ITER 2 (policy artifact only, no signed-OTA deployment at TRL 3)	Partial
I5	Insecure/ outdated compo-nents	4.2	—	ESP-IDF v5.4.2 (fixed build version, see README); no deprecated crypto primitives (ECDSA/SHA-256; AES-256-GCM via `mbedtls`)	Full

Table 19: OWASP IoT Top 10 coverage analysis (continued from Table 18) — Part 2 (I6–I10).

#	OWASP IoT Risk	Stage(s)	Phase(s)	Evidence in SMG	Coverage
I6	Insufficient privacy	1.4.3 + 2.2.4	P3	Data sovereignty policy (Stage 1.4.3): self-hosted RPi4; no PII; sensor data only; P3 encrypts all telemetry in transit with authenticated encryption	Full

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Table 19 continued from previous page

#	OWASP IoT Risk	Stage(s)	Phase(s)	Evidence in SMG	Coverage
I7	Insecure data transfer/storage	3.1.1 + 2.2.4	P3	P3 provides AES-256-GCM encryption in transit, now with forward secrecy (ephemeral ECDH, Table 3; hardware-validated, Section 6.3); no device-identity secret at rest. V_B and the WiFi credential are both unencrypted in flash (standard ESP32 practice; flash encryption is an ITER 2 candidate), so storage is not yet covered to the same standard as transfer	Partial
I8	Lack of device management	7.1	—	Stage 7.1 server records authentication status and session history in server-side logs (node_id, timestamps); heap and send-rate monitoring are specified but not yet implemented (Limitations); without a cryptographic chain, this is an operational log rather than an independently verifiable activity audit	Partial

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Table 19 continued from previous page

#	OWASP IoT Risk	Stage(s)	Phase(s)	Evidence in SMG	Coverage
I9	Insecure default settings	2.2.3 + 6.1.3	P1	No WiFi/server comms. until enrollment data exists in NVS; identity proof derives from the chip's physical PUF response, not a stored/default value; confirmed by cloning SMG_NODE_01's entire NVS onto SMG_NODE_02's physical chip, which the server correctly rejected (FT-06)	Full
I10	Lack of physical hardening	—	—	Out of scope for software methodology; addressed at org. level (Stage 1.4.1); JTAG disable and secure boot specified for TRL 5+	Partial
	Coverage score (unweighted)				**6/10 full + 4/10 partial**

Answer to CSQ4: Unweighted OWASP IoT Top 10 coverage of 6/10 items fully addressed and 4/10 partially addressed. Five of the six fully-covered items (I1–I3, I6, I9) have direct evidence from protocol phases P1–P3; the remaining one (I5: component vetting) is addressed by Stage 4 policy artifacts. The four partial items are I4 (secure update: policy defined, signed-OTA deferred to ITER 2), I7 (data transfer encrypted and forward-secret in transit, but V_B/WiFi credential unencrypted at rest pending ITER 2 flash encryption), I8 (device management: server-side logging without a verifiable chain, a trade-off for a smaller footprint—see Limitations), and I10 (physical hardening: outside a software methodology's scope). The coverage score is an unweighted count, not a severity-weighted metric (B12, Section 6.5).

6.5. Expert Validation

A structured questionnaire (14 Likert items, B1–B14, 5-point scale from 1 = strongly disagree to 5 = strongly agree, plus 3 open questions) was dis-

tributed to domain experts to supplement the case study evidence with independent professional judgment. Fifteen evaluators completed the questionnaire; fourteen consented to their responses being used as validation evidence (one declined consent and was excluded prior to analysis), exceeding the upper bound of the planned 6–10 evaluator target. Six requested anonymity and are referenced anonymously below. Evaluators held primary expertise in cybersecurity, IoT systems design/development, smart-grid/energy management, software engineering, or embedded systems, with 1–15 years of professional or research experience and academic backgrounds ranging from B.Sc./B.Eng. to Ph.D., from institutions in Mexico, Spain, Portugal, and Canada. Eight of the fourteen are affiliated with institutions unconnected to the author team (CINVESTAV, ESCOM-IPN, a national Instituto Tecnológico, ECOSUR, Universidad Iberoamericana, Universidade de Aveiro, University of New Brunswick), making the panel majority-independent; the remaining six are affiliated with the author institutions (INAOE, home institution; CEIT-BRTA, co-author affiliation).

Quantitative results: excluding B6, B8, B9, B10 (superseded-design items, below), the headline mean was 3.96/5 (SD 1.02, median 4.0, 140 responses); across all 14 items, 3.98/5 (SD 0.96). Item means ranged from 3.57 (B12) and 3.71 (B1) to 4.43 (B10) and 4.36 (B3, B14); none fell below the questionnaire's neutral midpoint. Among the retained items, B3 and B14 (4.36/5) rate highest; B7 (RoT-as-process abstraction, 4.29/5) rates third. The independent-evaluator subset (N=8) returned a 10-item mean of 3.90/5 (14-item: 3.89/5), modestly below the affiliated subset's 4.05/5 (14-item: 4.11/5); this mild affiliation skew is mitigated by the panel's majority-independent composition.

Qualitative findings that qualify the results: B12 (OWASP coverage sufficiency) returned the lowest item mean (3.57/5): two evaluators noted the 9/10 OWASP score (as presented) is an unweighted count, not severity-weighted, and does not establish that I10 (physical hardening) is the least critical omission; subsequent reclassification of I4, I7, and I8 to partial yields the revised 6/10 full + 4/10 partial reported in CSQ4 (Section 6.4). B1 (structure sufficiency) returned the second-lowest mean (3.71/5): three evaluators scored it 2/5, noting the methodology specifies *what* each stage must produce but not *how*—per-artifact templates or acceptance criteria would close this gap, echoed by several open-question (C3) responses. One evaluator qualified B6 (which, as presented, asked about the superseded 12-parameter model, note † above), noting the parameter count is an artifact

Table 20: Expert questionnaire Likert results (N=14; 1 = strongly disagree, 5 = strongly agree). SD is the population standard deviation across the 14 responses to that item.

Item	Statement (abbreviated)	Mean	SD	Mdn
B1	Four-phase, seven-stage structure gives sufficient guidance without external references	3.71	1.03	4.0
B2	Organizational phase (Stage 1) adds practical value ahead of architecture/technology decisions	3.79	1.26	4.0
B3	Stage-by-stage artifact requirements make correct methodology application verifiable	4.36	0.72	4.5
B4	TRL-guided iterative model with M1–M3 maintenance points is practical	3.86	0.83	4.0
B5	Entity-based identity model is clearer than role/policy-based approaches	3.86	0.99	4.0
B6†	12-parameter identity model generalizes beyond the SMG case study	3.86	0.74	4.0
B7	RoT-as-process abstraction accommodates heterogeneous device capabilities	4.29	0.80	4.5
B8†	Dynamic RoT execution (Process 6.1.3) is a verifiable identity mechanism	4.00	0.93	4.0
B9†	Six-phase authentication/data-exchange protocol (P1–P6) is logically complete	3.86	0.64	4.0
B10†	Hash-chain mechanism (P4/P6) gives meaningful, feasible integrity guarantees	4.43	0.62	4.5
B11	Technology-independent STRIDE mapping is more useful than technology-specific models	4.07	1.16	4.5
B12	90% OWASP IoT Top 10 coverage is sufficient security-by-design evidence at TRL 3	3.57	0.98	4.0
B13	Parameterized, technology-agnostic design applies beyond energy management	3.79	1.15	4.0
B14	More systematic security integration than existing IoT methodologies	4.36	0.72	4.5
	Full-panel mean (14 items, 196 responses)	**3.98**	0.96	4.0
	Headline mean, excl. † (10 items, 140 responses)	**3.96**	1.02	4.0

Note († B6, B8, B9, B10): these item texts, as presented to evaluators, name specifics of a superseded design – a 12-parameter identity model (B6), the superseded protocol's own RoT execution mechanism (B8, a random per-node key rather than a PUF response), a six-phase P1–P6 protocol (B9), and a hash-chain integrity mechanism (B10) – that the current implementation (Section 6.3) does not share: 14 parameters, a PUF-derived identity, phases P1–P3, no hash chain. B7, despite also sitting in Section 3.3 of the reference document, asks about treating the RoT as a process in general rather than about any mechanism-specific detail, so it is retained. The panel was not re-surveyed against the current implementation, so these four items are excluded from the headline mean reported elsewhere in this paper; the full-panel mean (including them) is reported here for completeness.

Note (B12): item text reflects OWASP coverage as presented to evaluators (9/10 full + 1/10 partial = 90%); the coverage assessment reported in CSQ4 (Section 6) is 6/10 full + 4/10 partial, since I4 (secure-update policy), I7 (data encrypted in transit but not at rest), and I8 (device management) are all classified as partial rather than full.

of the SMG's four entity types rather than a fixed model limit – an observation that still applies to the current 14-parameter model (Table 12), with generalization to other entity types untested. Two evaluators gave B13 (cross-domain generalizability) dissenting scores (2/5, 1/5), treating it as an unvalidated hypothesis (reflected in External Validity and Limitations below). One evaluator questioned whether same-team case-study evidence is sufficient without an independently-managed development team; the SMG implementation was carried out by multiple co-authors in distinct roles (Author Contributions), which partially addresses this. Two evaluators outside cybersecurity specifically (software-engineering methodology and general IoT systems) noted several protocol/threat-model items fell outside their expertise; their ratings are retained but should be weighted accordingly. The panel's majority-independent composition and subset means are reported above; the residual affiliation skew, convenience-sampling constraint, and preliminary-data caveat are acknowledged as limitations below.

7. Discussion

This work addresses two barriers to secure IoT ecosystem development: the security activation gap at the operational level, and the absence of a structured development methodology that integrates cybersecurity from the earliest organizational stage. The results of the confirmatory case study and expert validation are discussed below in relation to each research question.

CSQ1 — Methodology guidance: The stage-by-stage traceability (Tables 10 and 11) demonstrates that the methodology produces actionable decision points at every stage. The organizational phase (Stage 1) carries the highest leverage: classifying the SMG as critical infrastructure (Process 1.4.1) directly constrained the authentication scheme (requiring mutual cryptographic authentication), the storage approach (requiring isolation of identity material from firmware, met at TRL 3 by the NVS-partition/PUF design; a dedicated secure element is an ITER 2 requirement, see Limitations), and the access control granularity (requiring node, user, and admin distinction). Absent an explicit organizational stage, these constraints might not have been systematically propagated to architectural and implementation decisions.

CSQ2 — Identity model: The formal separation of identity into P_{pub}, P_{priv}, and P_{auth} subsets yielded a technology-agnostic security specification that translated directly to ESP32 NVS storage for the PUF helper data and

to an HKDF-derived ECDSA key pair for the private, non-persistent PUF response; `id_a`'s translation now also follows this path when provisioned (overriding the default compile-time placeholder with a per-device NVS value, see Limitations). The 14-parameter model produced zero ambiguity across 4 entity types, confirming the model's expressiveness for heterogeneous IoT ecosystems, including resource-constrained devices.

CSQ3 — Root of Trust verifiability: The $N = 30$ real hard-reset trial (Section 6.3) establishes that the handshake, including the forward-secrecy and helper-data-manipulation fixes, authenticates reliably (30/30, every trial on the first attempt) despite per-attempt PUF measurement noise, with the P3 demo message delivered and decrypted successfully every time – not merely that it can succeed once.

CSQ4 — Threat coverage: The unweighted OWASP IoT Top 10 coverage (6/10 full; 4/10 partial: I4 secure-update policy only, I7 data encrypted in transit but not at rest pending ITER 2 flash encryption, I8 device management without a verifiable chain, I10 physical hardening) indicates the methodology systematically addresses known IoT threat categories. The STRIDE analysis operates at two levels: the methodology-level mapping (Tables 5/6, Section 4.7) shows every threat addressed by a mechanism mandated by at least one stage; the protocol-level mapping (Tables 13–14, Section 5.6) traces each mechanism to its implementing phase. All six categories are addressed, four by multiple independent phases. The three partial OWASP items reflect two deliberate trade-offs (I4, I8—see Limitations) and one inherent scope boundary (I10); the coverage figure is unweighted, not severity-weighted.

Maintenance cycles: At least seven maintenance moments were triggered during Stages 5–6 (M2 firings for code-level revision, Section 6.3, Limitations), confirming the methodology's M2 pathway operates as specified at TRL 3. Each was caught during deployment-phase bench testing or subsequent review, before or alongside the reported trial; the case study does not establish a counterfactual about whether any would have surfaced only later absent that structure, since bench testing of this kind is standard practice regardless. M1 and M3 were not triggered and remain empirically unvalidated, available for future maintenance points and the operation-phase TRL transition.

7.1. Limitations

- **Single topology:** Only centralized topology was implemented at TRL 3. Decentralized and mesh topologies supported by Stage 2.1.4 remain to be

validated empirically.

- **TRL 3 scope:** The proof of concept operates in a controlled laboratory environment. Environmental variation, scalability beyond 2 nodes, and production-grade hardening (JTAG disable, secure boot) are deferred to ITER 2.

- **Primary-agent power dependency:** In the TRL 3 bench configuration the RPi4 primary agent is powered from the same DC bus it coordinates (via a DC/AC inverter); SSR duty transients and node resets cause primary-agent brownouts that can terminate sustained bench trials early. This coupling constrains sustained concurrent operation and is an artifact of the laboratory power arrangement rather than the methodology; ITER 2 will use an independent primary-agent supply.

- **Forward secrecy (ephemeral ECDH added, hardware-validated):** An earlier revision derived Key_{ab} from V_B and the handshake nonces alone, so recovering V_B (e.g., from a flash dump, Table 6) together with a recorded handshake sufficed to recompute that session's key. Key_{ab} now additionally depends on Z, a per-session ECDH secret discarded immediately after use, closing that gap: recovering V_B alone no longer yields Key_{ab} for any session. The fix is implemented (`rot_session.c`/ `rot_challenge_service.py`), passes the host-side protocol test suite (including a dedicated test that two handshakes against the same node yield different keys), and was validated end-to-end on the physical ESP32 against the real RPi4 server with a refreshed $N = 30$ bench round (30/30 success, Table 17, superseding the earlier pre-fix timing retained in the artifact repository for provenance).

- **Device-side PUF entropy:** The server stores only pk_A and V_B (Section 4), neither of which recreates sk_A if disclosed; however, sk_A's security still depends on the actual (not merely designed) entropy of the PUF response it derives from. Independent enrollments on one SMG node have measured 421–422-byte (∼3368–3376-bit) stable responses, somewhat above the library's ∼380–400-byte reference-hardware estimate (chip-to-chip and enrollment-to-enrollment variation, discussed below). Raw length is not min-entropy: repetition-code helper data is known to leak some response information (Dodis et al., 2004; Maes, 2013; Delvaux

et al., 2015), so 3368 raw bits do not imply 256 bits of adversarial min-entropy. A dedicated min-entropy assessment (e.g., NIST SP 800-90B) is the direct way to confirm sufficient min-entropy for the 256-bit scalar it seeds; this is a non-trivial statistical analysis of repeated-boot raw measurement data that this TRL 3 validation does not yet include, and is deferred to ITER 2 alongside the other hardening items in this list.

- **Two PUF/NVS defects found and fixed during this work (full discovery-and-fix history in Supplementary Material S2):** `esp32_puflib`'s enrollment procedure left a raw per-bit measurement blob (`PUF_FREQUENCY`) unerased in NVS, sufficient to reconstruct the PUF reference without the physical device, contrary to this paper's claim that no raw response is ever written to flash; the fix zero-overwrites and erases it post-enrollment, though NVS's log-structured storage means an earlier, un-zeroed version can still be byte-recoverable until garbage collection runs (SMG_NODE_01's deployed NVS carries none of this residue, as a side effect of how its server reference was provisioned, not a property the library guarantees in general). Separately, the public helper-data blobs (`PUF_MASK`/`ECC_DATA` and their deep-sleep-retry counterparts) were stored without integrity protection, letting an attacker with flash-write access use the ECC decoder's error count as an oracle on individual response bits; the fix binds SHA-256 of all four blobs into the HKDF key derivation (applied in two passes, after a second review found the first covered only two of the four blobs), confirmed load-bearing by a direct single-bit-flip test. Only flash encryption and secure boot (ITER 2 candidates, not yet enabled) close these fully. Only the second fix (the helper-data binding) changes the derived key pair and required re-enrolling SMG_NODE_01, in each of its two passes; the `PUF_FREQUENCY` erasure does not change the derived key (re-validation results below).

- **PUF library build and bench status (validated on both nodes):** `esp32_puflib` (Stanicek, 2022) is a native C/ESP-IDF component, linked directly into the node's firmware without a language-runtime wrapper; it and its RPi4 counterpart build cleanly and have been bench-validated end-to-end on both of the SMG's physical ESP32 units ($N = 30$ each, 30/30, Section 6.3). Physical distinctness across chips was additionally confirmed by cloning SMG_NODE_01's NVS onto SMG_NODE_02's hardware: rejected by the server in all 11 cross-chip attempts across four

successive firmware generations (Supplementary Material S2, Incident 4), since detection rests entirely on the server's ECDSA verification, not any device-side check – relaxing the device-side diagnostic only changes when, not whether, a wrong reconstruction is caught.

- **Sensor/EMS modules not yet implemented (C/ESP-IDF firmware scope):** The `smg_puf_rot_node` C/ESP-IDF firmware currently implements only the identity (PUF), Root of Trust, and secure-communication layers; fuzzy Energy Management System and sensor-acquisition logic have not yet been implemented in this firmware. A full end-to-end SMG deployment requires this additional work, distinct from and in addition to the hardware bench validation above. The same gap applies to the server side: the primary agent's anomaly-based trust scoring and quarantine logic (Stage 7.2.1) exists in `smg_primary_server`'s original module set (`auth_service.py`, `ems_coordinator.py`) but has not yet been ported to `rot_challenge_service.py`, the server counterpart of the current RoT protocol; DoS/I3 coverage claims elsewhere in this paper that cite trust scoring describe the methodology-mandated mechanism and its prior instantiation, not a property of the currently-deployed RoT server.

- **No cryptographic session-chain audit (deliberate design simplification):** The SMG's Root of Trust does not implement a per-message hash chain to provide a retrospectively-verifiable, third-party-auditable record of session integrity. Message-level integrity is provided by the AES-256-GCM authentication tag (P3), but there is no cryptographic chain binding messages within a session, and Repudiation/I8 (device management) coverage is therefore evidentiary (server-side session logs) rather than independently verifiable (Table 5, Table 19). Introducing a lightweight, chain-free sequencing mechanism (e.g., a monotonic per-session counter bound into the GCM associated data) without the overhead of a full chain audit is a candidate ITER 2 refinement if reviewers or subsequent evaluation identify this trade-off as unacceptable.

- **PUF reliability on the deployed hardware (cross-node data now available; thermal/voltage characterization pending):** `esp32_puflib` targets a design bit-flip probability of 0.001–0.005 via a stable-bit mask and 8×-repetition-code error correction, yielding an estimated ∼380–400-byte stable response in the library's own design notes; in-

dependent enrollments on SMG_NODE_01 have produced 421–422 bytes, consistently reproducible but somewhat above that estimate (a plausible chip-to-chip variation, not a conformance failure, since the library documents its figure as an estimate). Reconstruction reliability was measured over six $N = 30$ bench rounds on SMG_NODE_01 across the project's history (Table 17), including one under a since-relaxed reliability gate that rejected every trial (Supplementary Material S2, Incident 3); the round reported there shows no bimodal attempt pattern, with every trial succeeding on the first attempt – the two-attempt pattern seen in the 2026-10-13 round is now understood as an artifact of that gate, not a genuine circuit-level difference between the library's two reset depths. That round's attempt-1 error rate rose measurably across its own ∼5-minute duration, suggesting within-run thermal drift; SMG_NODE_01's canonical round, though shorter (∼3 minutes), shows the same upward trend (0.342% over its first ten trials versus 0.436% over its last ten; Spearman $\rho = 0.74$), without affecting first-attempt success. SMG_NODE_02, registered and bench-validated end-to-end once a per-device `id_a` provisioning gap was closed (above), produced a 420-byte stable response (two bytes shorter than SMG_NODE_01's 422) and comparable reliability under the current firmware build: $N = 30$, 30/30 success, every trial on the first attempt, attempt-1 raw error 0.403–0.531% (median 0.455%, modestly higher than SMG_NODE_01's canonical-round range of 0.308–0.485%, mean 0.397%) (raw logs and summary in the artifact repository at `case_study_SMG/Implementation/smg_puf_rot_node/bench_results_20261019_node02/`, Materials Availability). This confirms the reliability profile is not an artifact specific to one chip. Handshake timing is not reported here as a cross-chip comparison: SMG_NODE_02's firmware build was several days newer than SMG_NODE_01's canonical round, and a same-build regression check on SMG_NODE_01 itself (6 handshakes observed, of which only 1 raw log was archived) suggests build-to-build timing varies by roughly 250 ms independent of any code change relevant to the handshake path – large enough that neither a cross-build nor a same-build comparison at this evidence level supports a firm latency claim; identifying the cause is deferred to future work. Two items remain unmeasured: (a) sensitivity to temperature/supply-voltage variation in a controlled way, deferred to ITER 2; and (b) SMG_NODE_02's own thermal-drift trend over a longer run, not separately characterized here.

- **Single application domain:** Energy management is the only domain tested. The expert questionnaire (N=14, Section 6.5) supports the technology-agnostic design intent (B13 mean 3.79/5) but also recorded two dissenting scores on this same item (2/5 and 1/5), with the dissenting evaluators characterizing cross-domain generalizability as an unvalidated hypothesis; empirical validation in additional domains remains future work.

- **Physical hardening:** I10 (physical hardening) and hardware-based Root of Trust (e.g., ATECC608 secure element) are not addressed at TRL 3 but are specified as ITER 2 requirements.

- **Identity lifecycle (methodology gap):** The methodology as presented does not specify a complete identity lifecycle process (parameter rotation, revocation, recovery after compromise without full re-provisioning, ownership transfer, and decommissioning). This is a gap in the methodology itself—not solely a TRL 3 scope limitation—since recovery from a compromised node currently requires re-running the full Root of Trust. An expert evaluator (Section 6.5) identified this as the single addition with the greatest potential impact on practical applicability; it is the highest-priority extension for the next methodology revision and is deferred to ITER 2.

- **User/administrator password hashing:** Users and administrators authenticate via a plain SHA-256 hash of their password (Table 12), not a salted, slow key-derivation function (bcrypt, scrypt, Argon2) — weaker than current practice, and a candidate ITER 2 hardening item; it does not affect the PUF-based IoT node scheme, which uses no password.

- **WiFi credential provisioning (fixed; fleet-sharing residual):** An earlier firmware revision compiled the WiFi SSID/password into source as a fixed constant, identical for every build and visible in the published firmware source. The credential is now provisioned into NVS out-of-band by `tools/provision_server_ref.py` (the same mechanism already used for V_B), so no WiFi credential value appears in firmware source or the compiled binary. One property is unchanged by this fix and is not a firmware defect: every node on the fleet's WPA2-PSK access point necessarily shares the same passphrase, so recovering one node's NVS still discloses the credential needed to join that network, same as recovering any

WPA2-PSK passphrase would. Rotating it still requires re-provisioning every node's NVS image, not merely one.

- **Per-device node identifier (fixed; found while registering a second node):** `id_a`, the device's wire identifier and server-registry key, was a compile-time placeholder shared by every unit built from the same firmware source, which blocked registering a second physical node under this scheme at all: the server registry is keyed on `id_a`, so a second node with the same value would have overwritten the first's registry entry. `rot_identity_get_id_a()` now reads `rot_identity/id_a` from NVS if present, falling back to the compile-time placeholder otherwise, so each node can be provisioned with a distinct value (`tools/provision_server_ref.py -id-a`) without a firmware change; `id_a` is a public value bound into the signed P2 transcript (`rot_session.c`), not the secret the proof rests on (the PUF-derived key pair), so this fix does not change what is cryptographically proved, only registry addressing. The fallback was verified not to disturb SMG_NODE_01 (6/6 handshakes against its existing registry entry, byte-identical before and after the firmware update); SMG_NODE_02 was then provisioned with a distinct `id_a` and validated end-to-end (Section 6.3).

- **Expert validation evaluated an earlier instantiation, not the final PUF-based system (scope of what the panel's ratings establish):** The reference document distributed to evaluators (June 24, 2026) does not merely use more conservative figures than the final implementation; it describes a materially different instantiation of the same methodology: three ESP32 nodes (not two), MicroPython firmware (not C/ESP-IDF), a random per-node key plus HKDF-SHA256 shared secret for RoT execution (not a PUF-derived identity), a continuous hash-chain integrity mechanism (not present in the current protocol), and a 12-parameter identity model (not 14). Four questionnaire items name these specifics directly (B6, B8, B9, B10) and are excluded from the headline mean for that reason (Section 6.5); the retained 10 items ask about the methodology's process-level properties – stage structure, artifact verifiability, the public/private/authentication identity taxonomy, technology-agnostic STRIDE mapping, cross-domain applicability – which both instantiations share, since they concern the methodology prescribing a step,

not which cryptographic primitive a given case study chose to satisfy it. The panel's quantitative ratings should accordingly be read as evaluating the methodology's conceptual soundness and process structure, as demonstrated through a representative (if since-superseded) instantiation; they are not a validation of the final PUF-based implementation's specific technical choices, which the $N = 30$ hardware bench and functional tests (Section 6.3) validate separately and are not contingent on the panel's scope. The panel was not re-surveyed against the current implementation; doing so is deferred to future work. Fifteen evaluators completed the questionnaire (14 Likert items, 3 open questions); fourteen consented to their responses being used as validation evidence (one declined and was excluded), exceeding the planned 6–10 target, yielding a headline agreement mean of 3.96/5 (10 items excluding the four superseded-design items above; 3.98/5 across all 14, Section 6.5). Eight of the fourteen are affiliated with institutions unconnected to the author team (majority-independent), though recruitment remained convenience-sampled; the independent subset (3.90/5) rated modestly below the affiliated subset (4.05/5), a mild skew the panel composition mitigates. At least two affiliated evaluators (Acknowledgements) are MSc students in the authors' own department, a power relationship beyond simple affiliation this study does not separately quantify.

7.2. Threats to Validity

8. Conclusions

This work has presented a security-by-design methodology for centralized IoT ecosystems that addresses the security activation gap by integrating cybersecurity systematically from the earliest organizational stage. The methodology's four phases and seven stages produce documented, traceable artifacts; the entity-centric identity model provides technology-agnostic security specifications applicable to heterogeneous devices; and the dynamic Root of Trust process establishes verifiable cryptographic identity before any operational communication begins.

Empirical validation through a confirmatory case study (Yin, 2014) applying the methodology to a DC Smart Microgrid IoT ecosystem yields four findings. First, all seven stages produced complete, traceable outputs with zero external guidance required (CSQ1). Second, the 14-parameter identity model across four entity types produced zero ambiguity (CSQ2). Third,

Table 21: Threats to validity.

Type	**Specific Threat**	**Mitigation**
Internal validity	SMG decisions may reflect author preference rather than methodology guidance; the traceability tables are an authored narrative and cannot fully rule this out on their own	Stage-by-stage traceability (Tables 10 and 11) documents an explicit process → decision chain; an independent code-level audit (AI-assistance declaration) cross-checked specific claims against the firmware/server source and forced correction of several that did not hold (e.g. the device-identifier and session-key claims in Limitations) — evidence against purely self-serving narrative, though not a substitute for independent replication
External validity	Single domain (energy) + single topology (centralized) limits generalizability	Methodology is technology-agnostic by design; expert questionnaire (N=14, headline mean 3.96/5, Section 6.5) supports this design intent, though three evaluators noted that generalization beyond the validated domain and entity set remains an untested hypothesis
Construct validity	TRL 3 does not demonstrate production readiness	TRL 3 is the stated objective of ITER 1; the methodology explicitly supports iterative maturation toward TRL 5+
Reliability	Physical prototype measurements subject to environmental variation	$N = 30$ real hard-reset trials on each of the SMG's two physical nodes (Section 6.3): handshake latency std 48 ms, end-to-end success 30/30 on both; ambient laboratory conditions (no controlled temperature/voltage sweep) remain a disclosed limitation

the RoT process established verifiable mutual authentication: a PUF-based scheme (`esp32_puflib` (Stanicek, 2022)) that reconstructs each node's identity from non-persistent physical characteristics rather than a stored key, combined with an asymmetric (ECDSA) device-authentication proof, a per-node symmetric server-authentication proof, and an ephemeral ECDH exchange giving the session key forward secrecy, validated end-to-end on real hardware for both nodes: SMG_NODE_01's detailed $N = 30$ hard-reset trial (30/30 success, every trial on the first attempt, median handshake latency 1300 ms, mean 1313 ± 48 ms) is the headline round reported, and SMG_NODE_02's independent $N = 30$ trial (30/30, also every trial on the first attempt) confirms the reliability profile generalizes across chips; this validation also found and fixed a miscalibrated local reliability gate that had originally rejected every attempt outright, a raw PUF measurement left unerased in NVS, a helper-data-manipulation oracle that an earlier form of the mitigation did not fully close, and a per-device node-identifier gap that blocked registering a second node at all (Limitations); only the helper-data-manipulation fix changes the derived key and required re-enrolling the device, in each of its two passes (CSQ3, Section 6.3). Fourth, the methodology achieved unweighted OWASP IoT Top 10 coverage of 6/10 items fully addressed and 4/10 partially addressed (I4 secure-update policy only; I7 data encrypted in transit but not yet at rest; I8 device management, a trade-off for a smaller on-device footprint; I10 physical hardening outside software scope) (CSQ4). This case study evidence is supplemented by an expert questionnaire (N=14, Section 6.5) that returned a headline agreement mean of 3.96/5 (10 items; 3.98/5 across all 14), with the strongest agreement on methodology completeness and systematic security integration (B3, B14 means 4.36/5) and qualified agreement on cross-domain generalizability and threat-coverage weighting.

The methodology's parameterized, technology-agnostic structure is designed to support application across diverse IoT domains; cross-domain generalization is a design intent supported by expert opinion (B13 mean 3.79/5, with dissenting 2/5 and 1/5 ratings characterizing it as an unvalidated hypothesis) rather than a demonstrated property, and empirical validation in additional domains remains future work. Future work will extend validation to: ITER 2 (TRL 5+ with hardware security elements and signed-OTA implementation), cross-domain deployment (healthcare, industrial automation), an expanded expert evaluator pool with improved institutional independence, formal cryptographic protocol verification of the SMG's PUF-based authentication scheme, resolution of the identity-lifecycle gap (rotation, revocation,

recovery), evaluation of per-node verifier wrapping or HSM-backed server storage to reduce central-server single-point-of-failure risk, and a lightweight chain-free sequencing mechanism to recover stronger Repudiation/I8 coverage without the on-device footprint of a full cryptographic session-chain audit.

Declarations

Ethics Statement

The expert-validation questionnaire (Section 6.5) collected professional opinions from domain-expert volunteers; it did not collect sensitive personal data, and participation required explicit informed consent (item D1) covering use of responses as validation evidence, with an option for anonymous participation and the right to decline (exercised by one respondent, excluded prior to analysis). Formal institutional ethics-committee review was not sought for this questionnaire: it solicited professional opinions on a published methodology from consenting domain-expert volunteers acting in their professional capacity, collected no sensitive personal data, and involved no identifiable risk to participants beyond the attribution they explicitly consented to (or declined) in item D1 – a category of low-risk, opinion-based survey research that falls outside the scope requiring committee review under the authors' home institution's research-ethics policy.

Materials Availability

The ESP32 firmware (`smg_puf_rot_node/`) and primary agent software (`smg_primary_server/`) are available under Apache 2.0 license (the original code in this work) in the companion code repository (Campos-Sanchez et al., 2026b): https://github.com/RazielCS/smg-secure-microgrid, archived with a permanent identifier at https://doi.org/10.5281/zenodo.23505100; the vendored `esp32_puflib` component (`components/esp32_puflib/`) retains its own upstream MIT license, as stated in that directory. Within `smg_primary_server/`, only `puf_server.py` and `rot_challenge_service.py` ran during the reported bench trials; the directory also retains an earlier, superseded module set (`auth_service.py`, `server.py`, `pc_client.py`, `protocol.py`, `crypto_utils.py`, `ems_coordinator.py`), kept because part of its functionality has not yet been ported (see Limitations) and clearly labeled as such in the repository's README. The complete research

artifacts — manuscript source, case-study documentation, raw bench-trial measurements (canonical CSVs and serial logs under `case study SMG/Implementation/smg_puf_rot_node/`), and the expert-validation instruments — are available under CC BY 4.0 license in the companion artifact repository (Campos-Sanchez et al., 2026c): https://github.com/RazielCS/smg-secure-microgrid-paper-artifacts. Device- and server-side secrets (the real PUF verifiers, NVS images, and WiFi credentials) are excluded from both repositories: provisioning files use sanitized `.example.json` templates with zeroed values, and the real WiFi credentials are replaced by the placeholder `CHANGE_ME_WIFI_PASSWORD` wherever they would otherwise appear (credentials are provisioned into device NVS out-of-band, never committed as firmware source, Section 4); see each repository's README for the exact exclusion list.

Data Statement

The raw bench-trial data underlying this work's quantitative RoT findings — the $N = 30$ hard-reset trial's per-run serial logs and parsed summary (Table 17), the server-side log captured during that run, the FT-06 cross-chip cloning negative test's device/server logs, the single-bit-flip tamper test's logs (FT-05), and the re-enrollment serial logs with read-only NVS-dump summaries (secret values redacted) — have been deposited in, and are cited and linked from, the companion artifact repository (Campos-Sanchez et al., 2026c). The expert-validation instrument (questionnaire text) is deposited there as well; the individual evaluators' raw responses are not, since several evaluators requested anonymity (Acknowledgements) and consent was not obtained to publish raw response-level data. No packet captures or stage-artifact documents (risk register, access-control matrix, etc.) beyond what is reported in the manuscript and supplementary material were produced as separate deposit-ready files. The device/server secrets noted under Materials Availability are excluded for security rather than confidentiality reasons and are not needed to reproduce or verify any reported result.

Competing Interests

The authors declare no competing financial interests or personal relationships that could have appeared to influence this work.

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Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used Claude (Anthropic, Claude Code) as an AI-assisted software engineering and writing-support tool for: implementing, debugging, and bench-testing the case study's firmware and server software; statistical analysis of the real-hardware bench-trial data reported in Section 6.3; drafting and revising manuscript text, including cross-section consistency verification; and preparing the companion code and artifact repositories cited in Materials Availability. An independent AI coding assistant (OpenCode, running third-party models) was additionally used to perform an independent critical-review pass of the manuscript, whose findings the authors evaluated and selectively incorporated. After using these tools, the authors reviewed, verified, and edited all AI-assisted output and take full responsibility for the content of the published article. AI tools are not listed as authors or co-authors of this work.

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Author Contributions

R. Campos-Sanchez: Conceptualization, Investigation, Formal analysis, Software, Writing — original draft. **L. Hernandez-Martinez:** Supervision, Validation, Writing — review & editing. **C. Feregrino-Uribe:** Supervision, Validation, Writing — review & editing. **A. Penuelas-Angulo:** Conceptualization, Formal analysis, Writing — original draft. **D. Cruz-Aguilar:** Conceptualization, Validation, Software, Writing — review & editing.

Vitae

[TODO: author biographies and photographs pending]

A short (maximum 100-word) biography and a passport-type photograph for each of the 5 authors — R. Campos-Sanchez, L. Hernandez-Martinez, C. Feregrino-Uribe, A. Penuelas-Angulo, D. Cruz-Aguilar — are to be supplied by each author before submission; photographs are to be submitted as separate figure files, not embedded here.

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